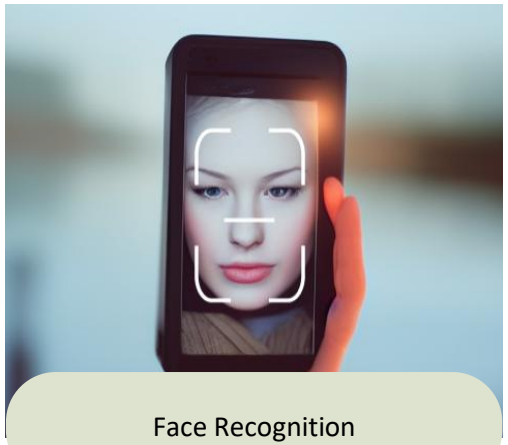
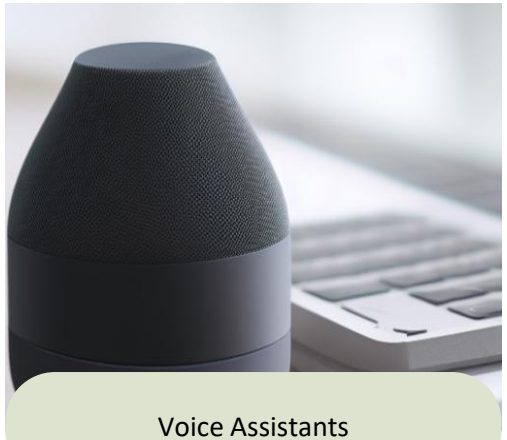
Artificial Intelligence Introduction

Prof. Dr. Raphael Herding

AI is Everywhere



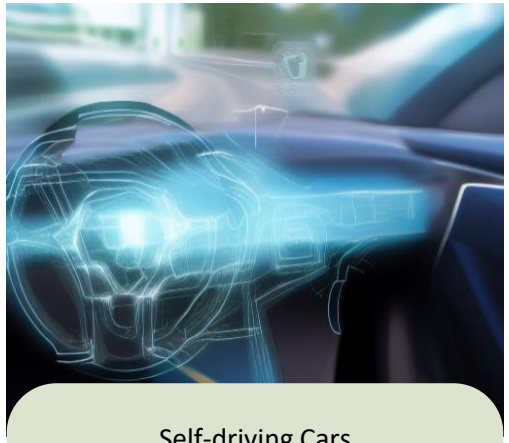
Face Recognition



Voice Assistants



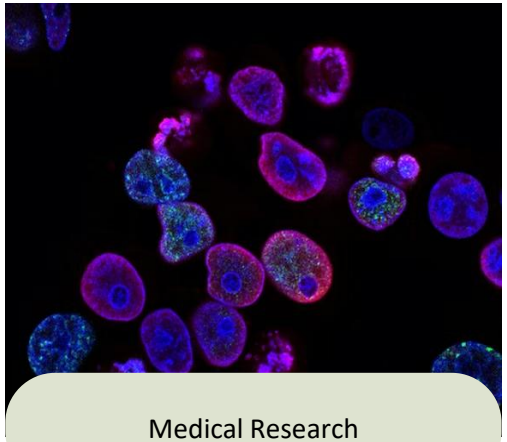
Product Recommendation



Self-driving Cars



Supply Chain Management



Medical Research



Fraud Detection



Gaming

Introduction



Introduction

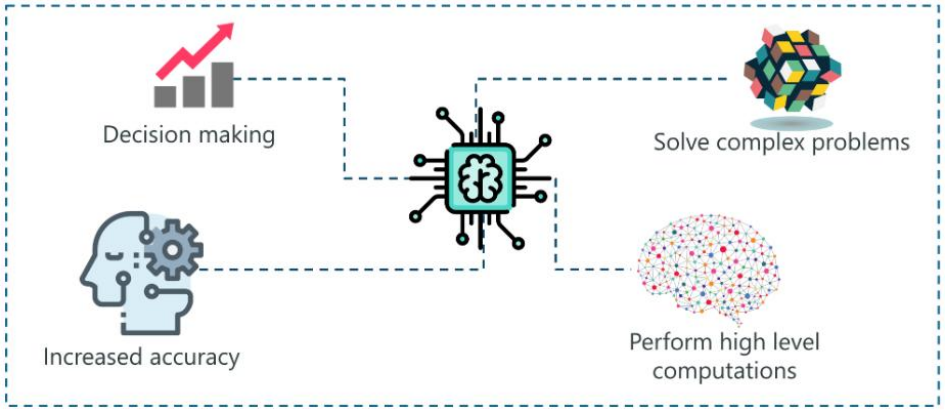
- Many new and old terms exists in the context of AI.
  - Machine Learning, Data Science, Data Mining, Data Analysis, Statistical Learning, Knowledge Discovery in Databases, Pattern Discovery, Big Data, …
- New challenges rises as well.
  - AI and Security, AI and Data Privacy, AI and hardware requirements, ethical challenges, …



Introduction

- In 1956, the term Artificial Intelligence was defined by John McCarthy. He defined AI as:

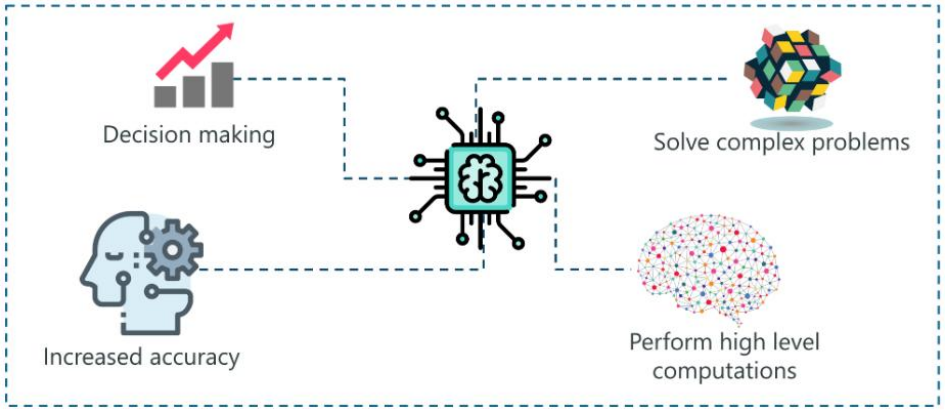
“The science and engineering of making intelligent machines.”



Introduction

- In 1956, the term Artificial Intelligence was defined by John McCarthy. He defined AI as:

“The science and engineering of making intelligent machines.”

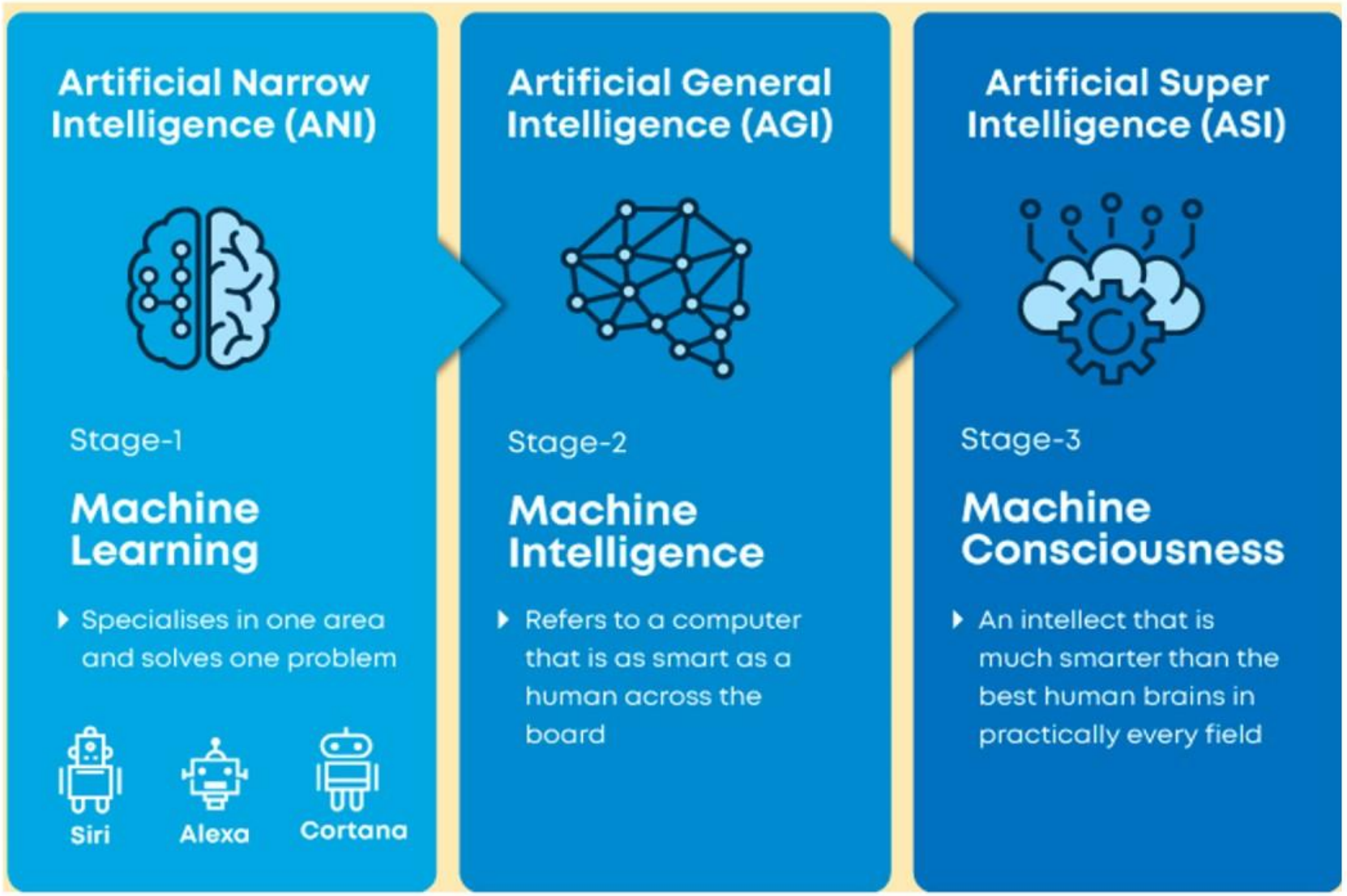


- Can be defined as the development of computer systems that are capable of performing tasks that require human intelligence.

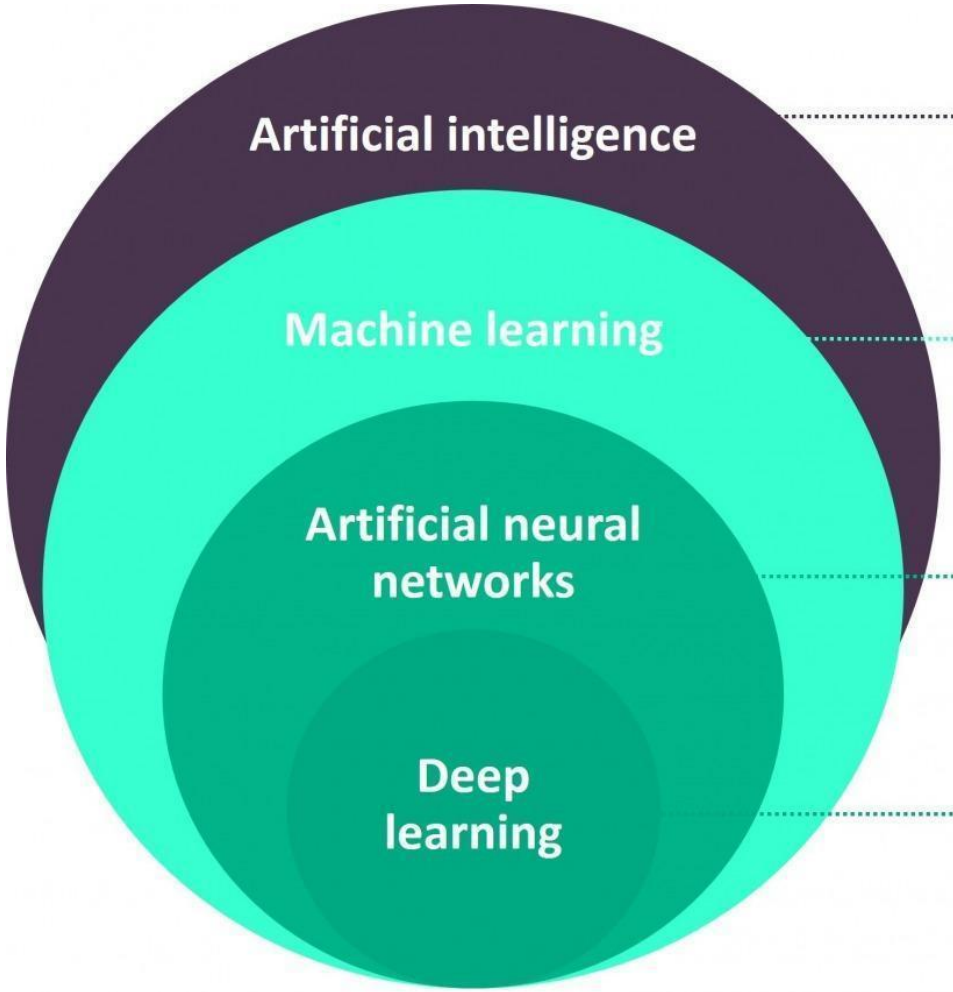
Introduction
Why Now?



Introduction
Stages Of Artificial Intelligence



Introduction
Disciplines of AI



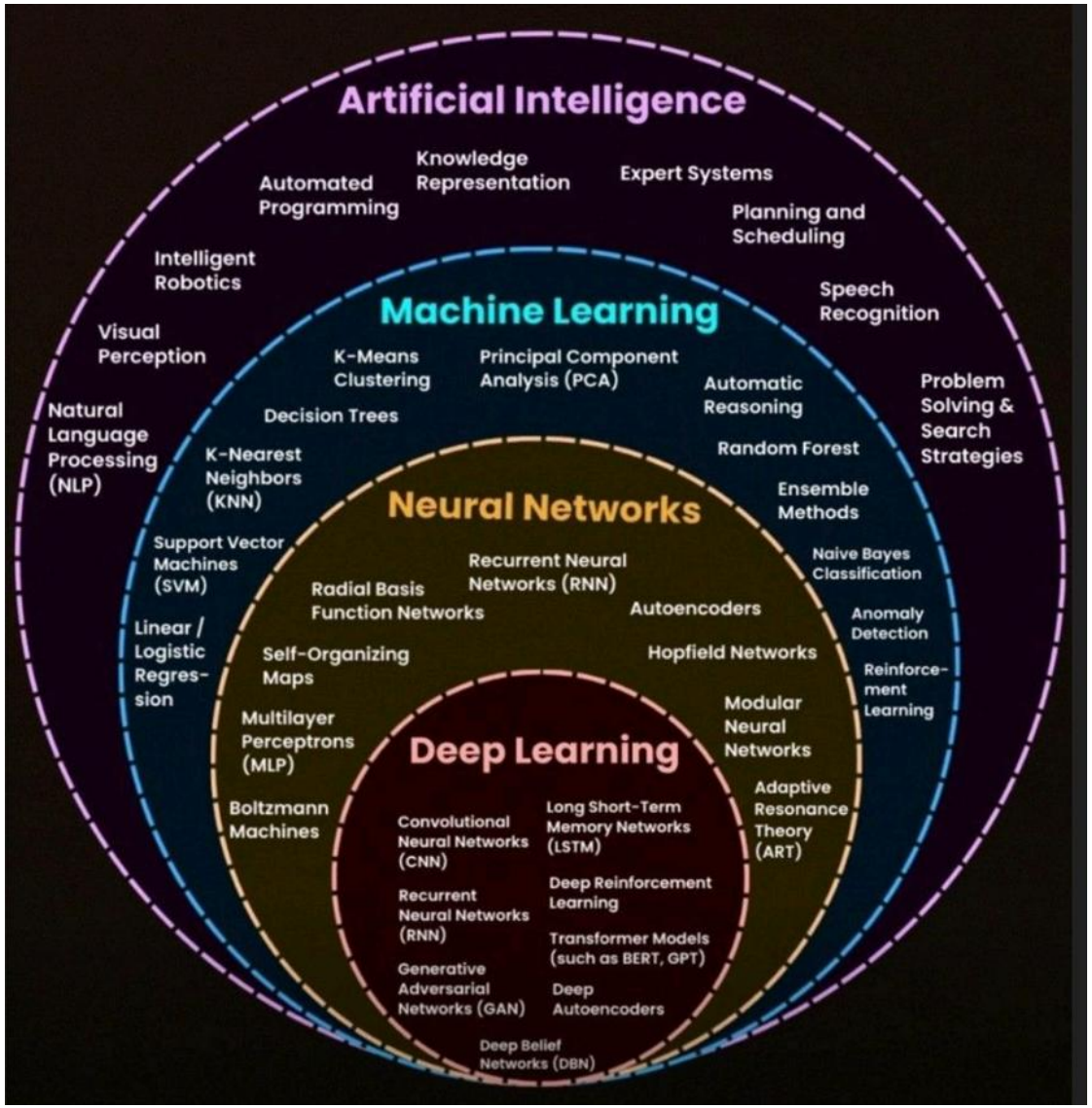
Artificial intelligence (AI)
Any techniques that enable machines to solve a task in a way like humans do

Machine learning (ML)
Algorithms that allow computers to learn from examples without being explicitly programmed

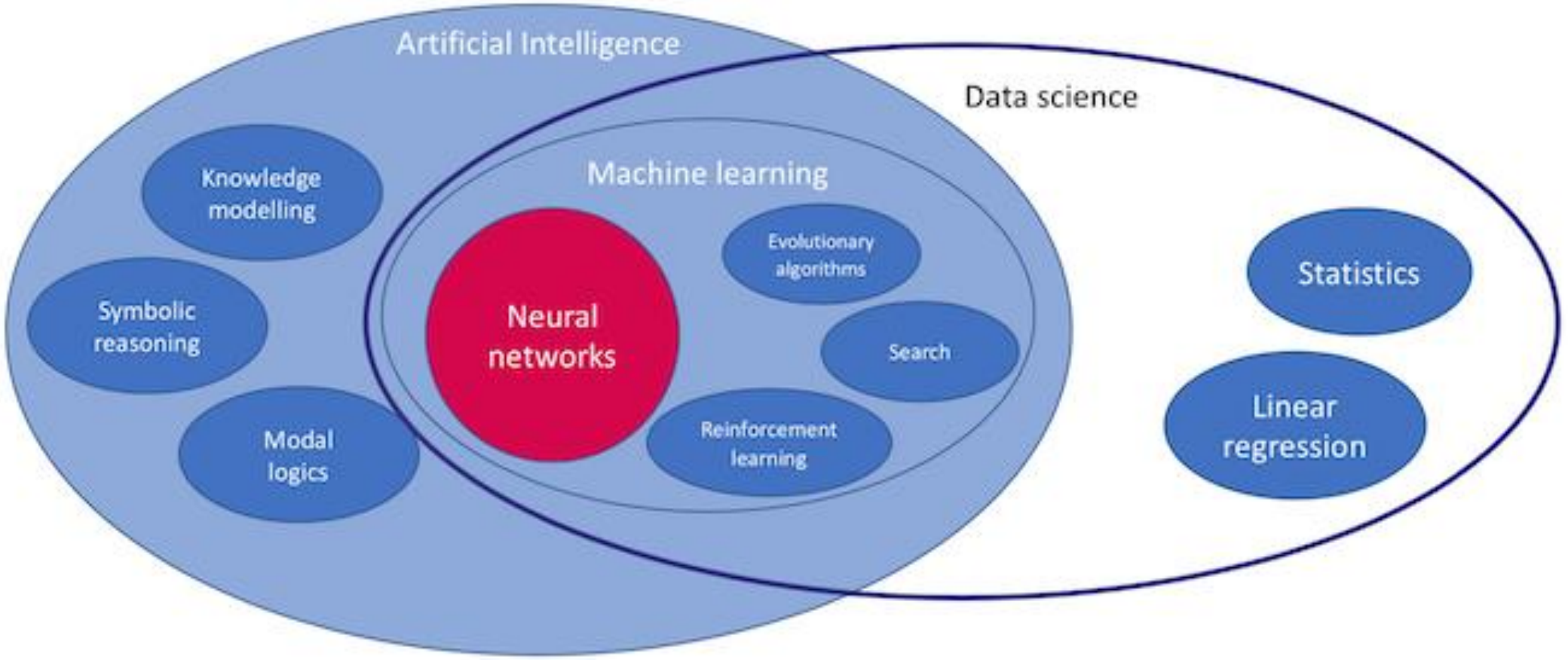
Artificial neural networks (ANN)
Brain-inspired machine learning models

Deep learning (DL)
A subset of ML which uses deep artificial neural networks as models and automatically builds a hierarchy of data representations

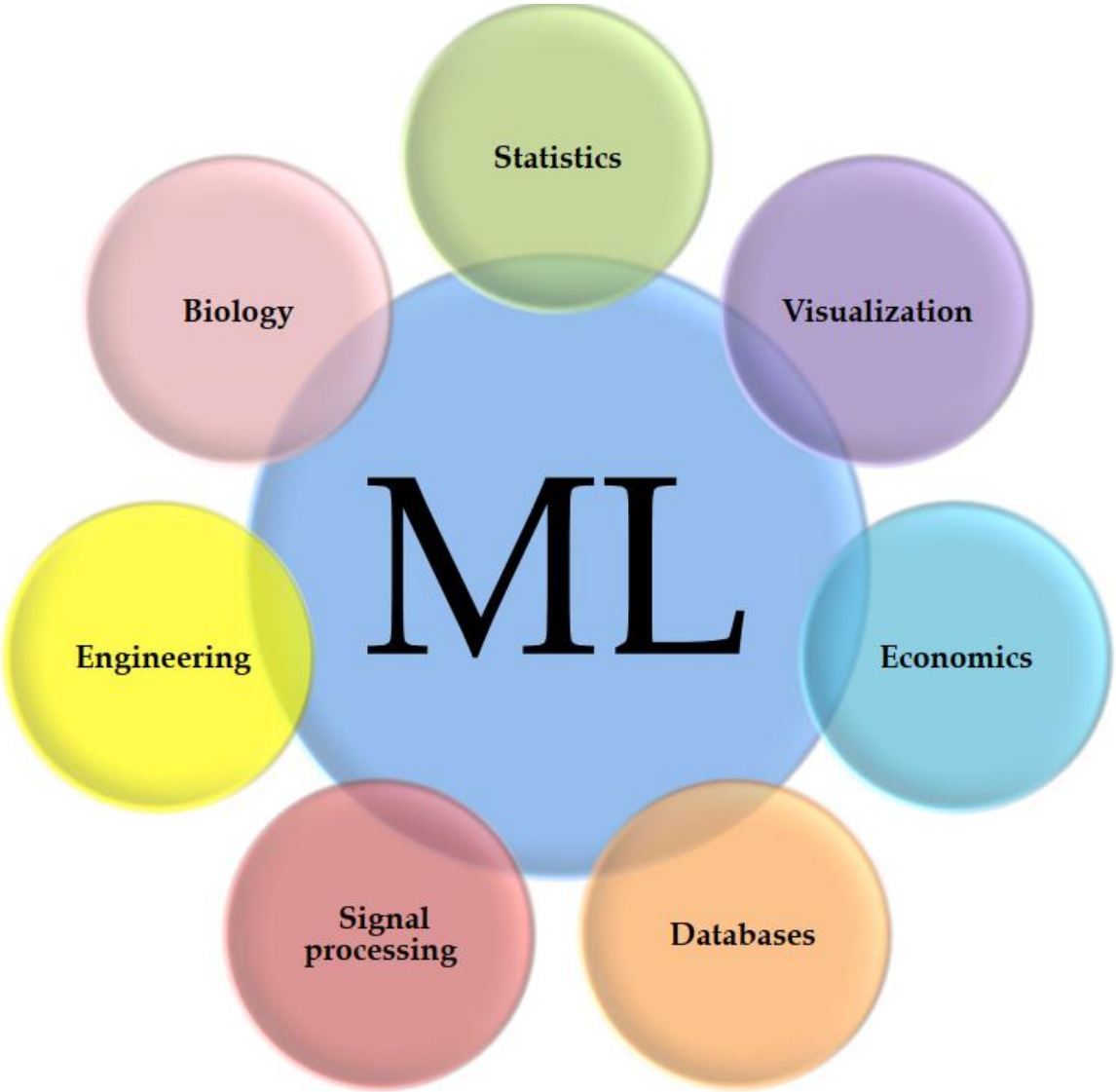
Introduction
Disciplines of AI



Introduction
AI and Data Science



Introduction
ML and AI is an Interdisciplinary Field



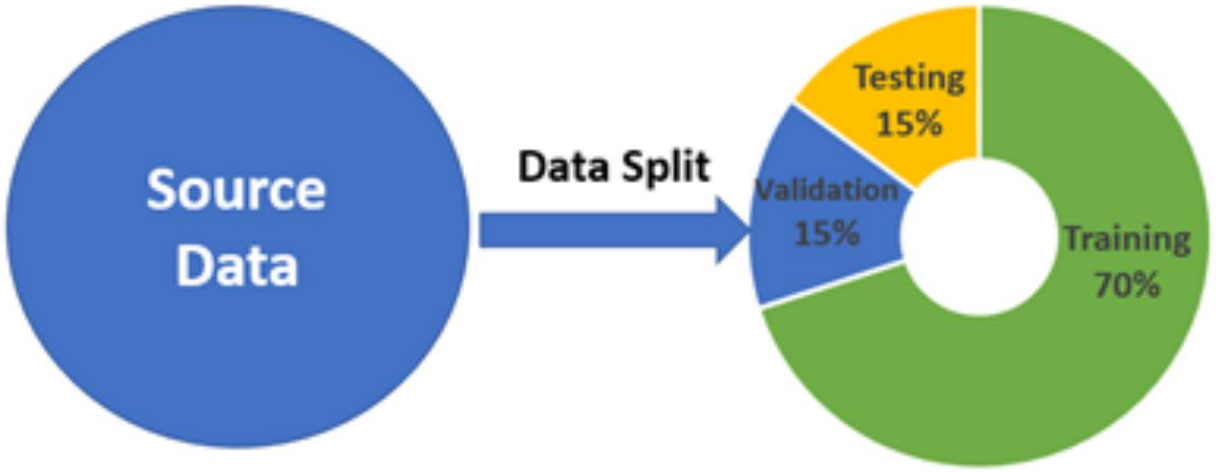
Introduction
Definition Learning of Software

"A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E. "

Tom Mitchell. Machine Learning 1997.

Introduction
Training and Testing

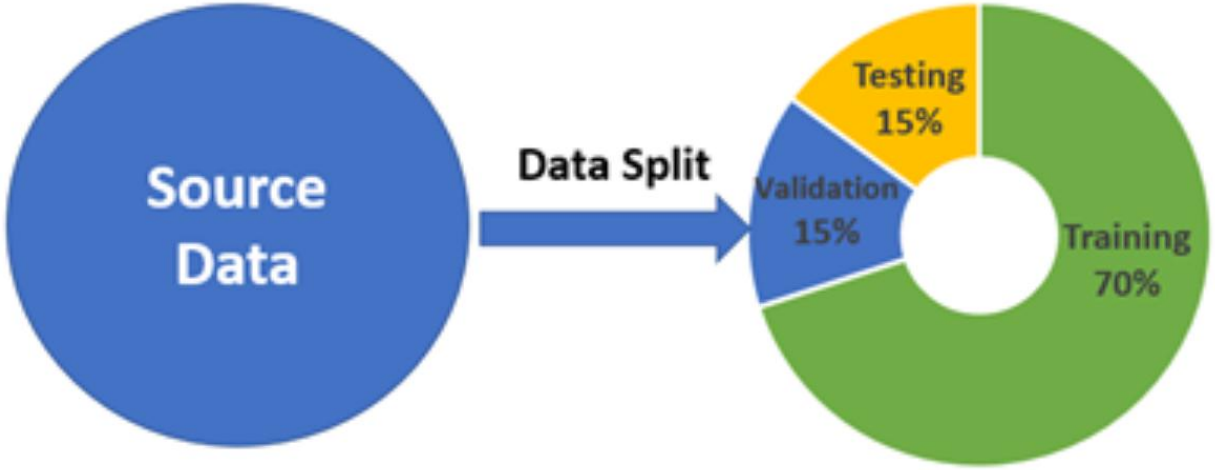
- Training is the process of making the system able to learn.
- Training, testing and validation set come from the same source.



- **Training Set**: It is the set of data that is used to train and make the model learn.
- **Validation Set**: Is used to validate the model performance during training.
- **Test Set**: Is used to test the model after completing the training.

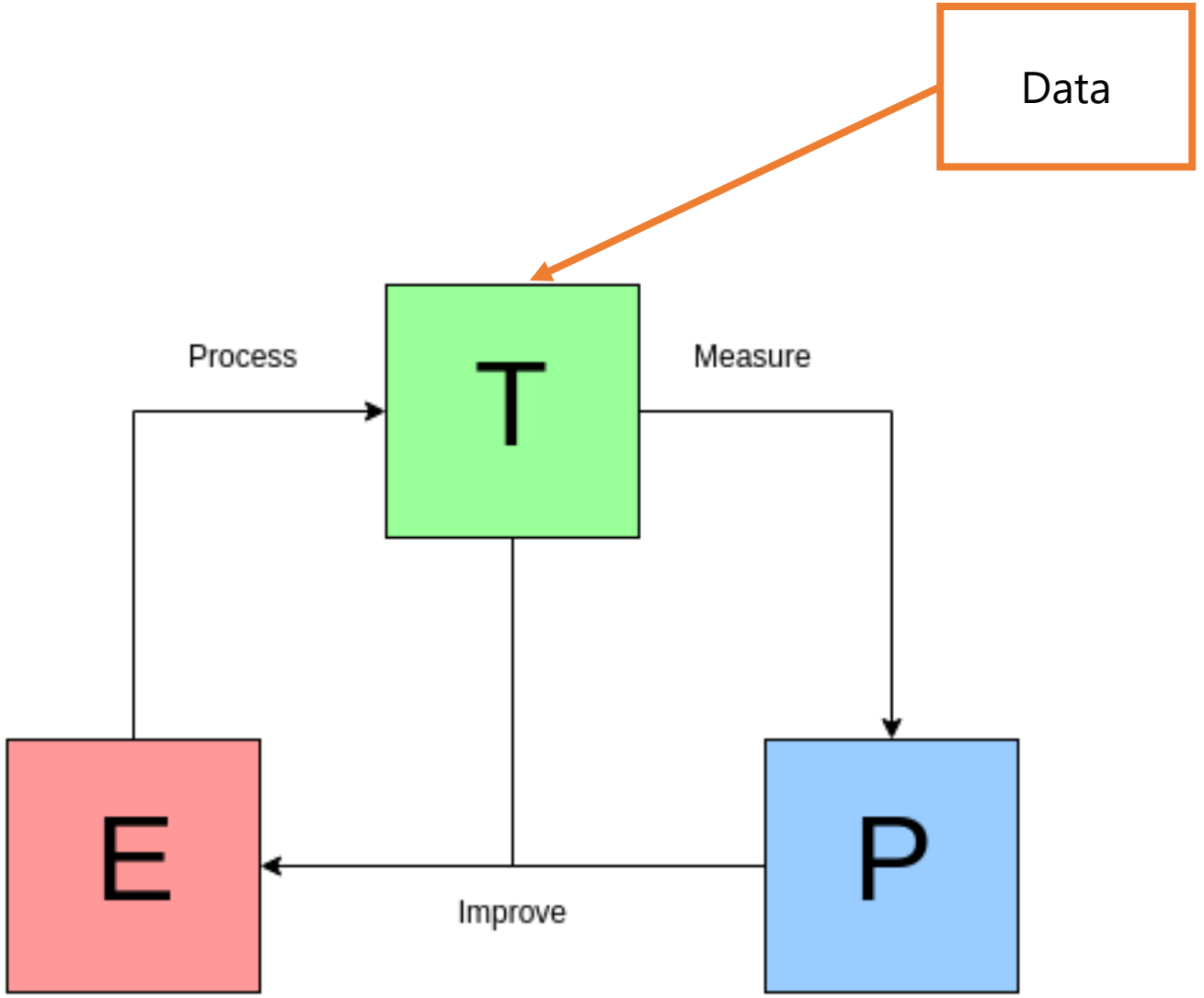
Introduction
Training and Testing

- Training is the process of making the system able to learn.
- Training, testing and validation set come from the same source.

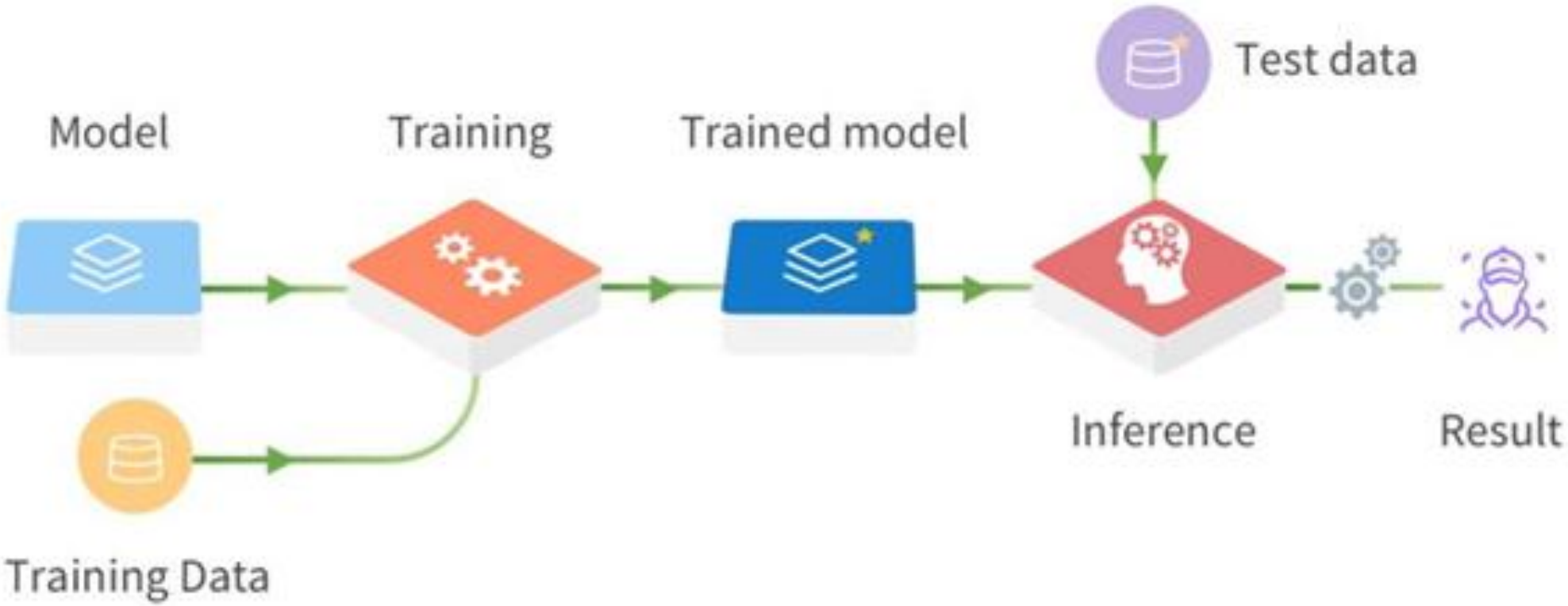


- **Training Set**: It is the set of data that is used to train and make the model learn. (Classroom teaching)
- **Validation Set**: Is used to validate the model performance during training. (Practice example problems)
- **Test Set**: Is used to test the model after completing the training. (Exam questions)

Introduction
Model Training (simple illustration)

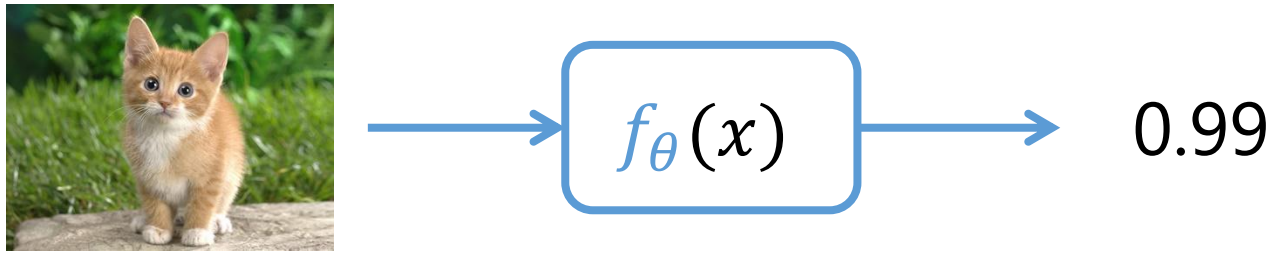


Introduction
Training and Testing Process

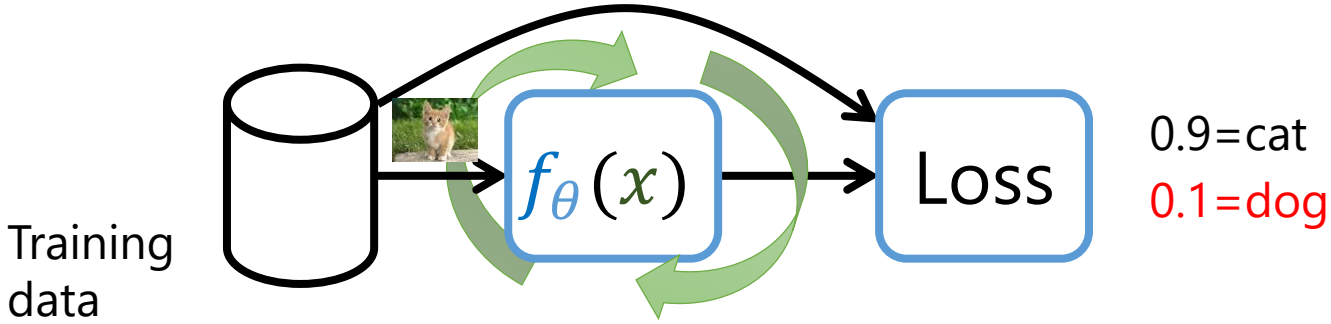


Introduction
Model

- **Model**: a physical, mathematical, or otherwise logical representation of a system, entity, phenomenon, or process (DoD 1998).
- A model in AI is a mathematical **function** specified by a set of **parameters.**



- **Training** tries to minimize the loss (error) of the function.



Machine Learning
Introduction

- Coined by American computer scientist Arthur Samuel in 1959, the term “machine learning” is defined as a

  computer’s ability to learn without being explicitly programmed.

- Machine learning uses programmed algorithms that receive and analyze input data to predict output values within an acceptable range.

- As new data is fed to these algorithms, they learn and optimize their operations to improve performance, developing “*intelligence*” over time.

Machine Learning
Types

Supervised Learning

- Makes machine Learn explicitly
- Data with clearly defined output is given
- Direct feedback is given
- Predicts outcome/future
- Resolves classification and regression problems



Unsupervised Learning

- Machine understands the data (Identifies patterns/structures)
- Evaluation is qualitative or indirect
- Does not predict/find anything specific

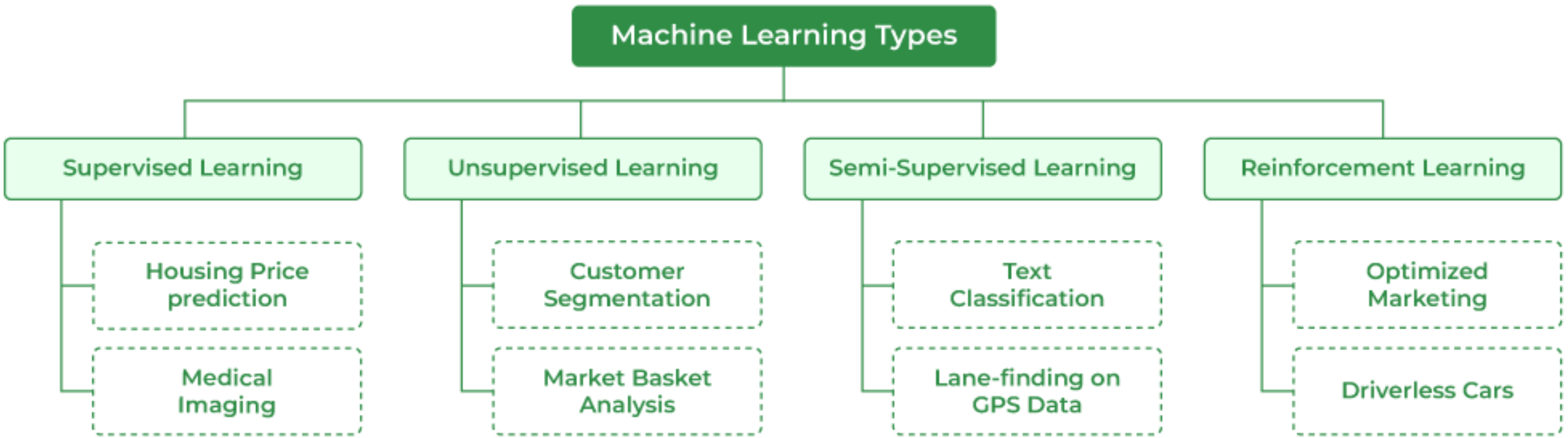


Reinforcement Learning

- An approach to AI
- Reward based learning
- Learning form +ve & +ve reinforcement
- Machine Learns how to act in a certain environment
- To maximize rewards



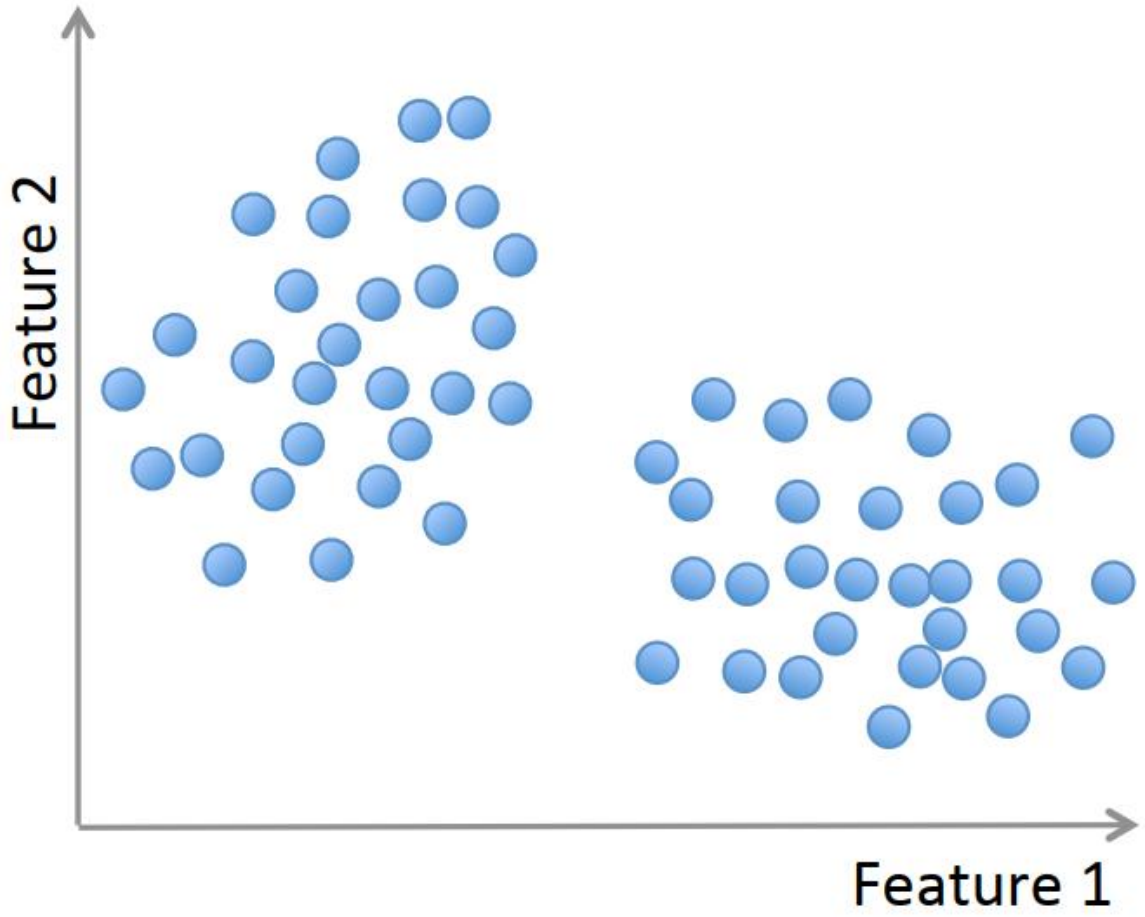
Machine Learning
Types



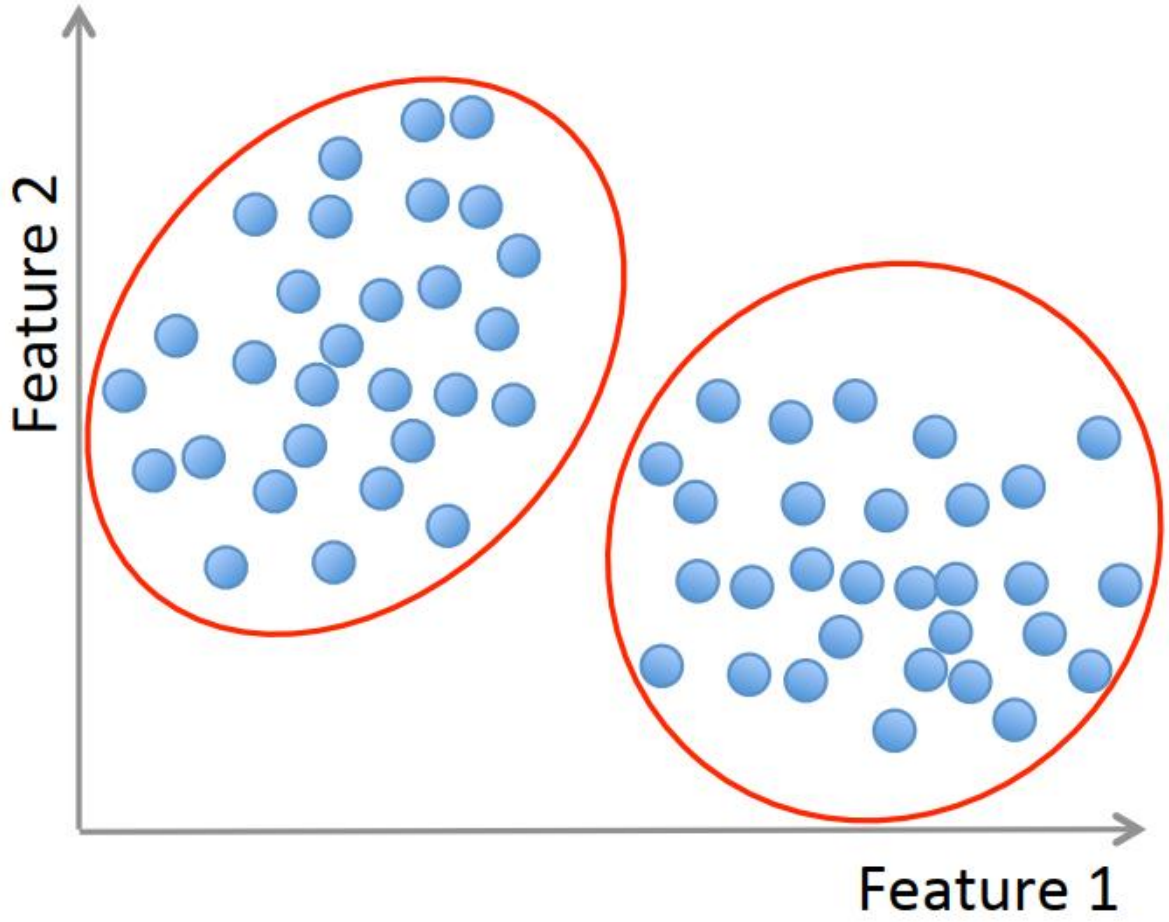
Machine Learning
Fundamental Techniques

- **Classification**: Unraveling Unknown Data Points
  - Is a type of supervised learning.
  - Goal is to predict the categorical label of new observations, based on past observations.
  - Assigning items into predefined categories or classes.
- **Regression**: Predicting Real Values
  - Is a type of supervised learning
  - Goal is to predict a continuous numeric value based on input features.
  - In contrast to classification, which assigns items to discrete categories, regression deals with predicting quantities.
- **Clustering**: Unveiling Data Similarities
  - Is a type of unsupervised learning.
  - Goal is to group a set of objects in such a way that objects in the same group (or cluster) are more similar to each other than to those in other groups.
  - It seeks to discover the inherent structure in the data (not labeled data).

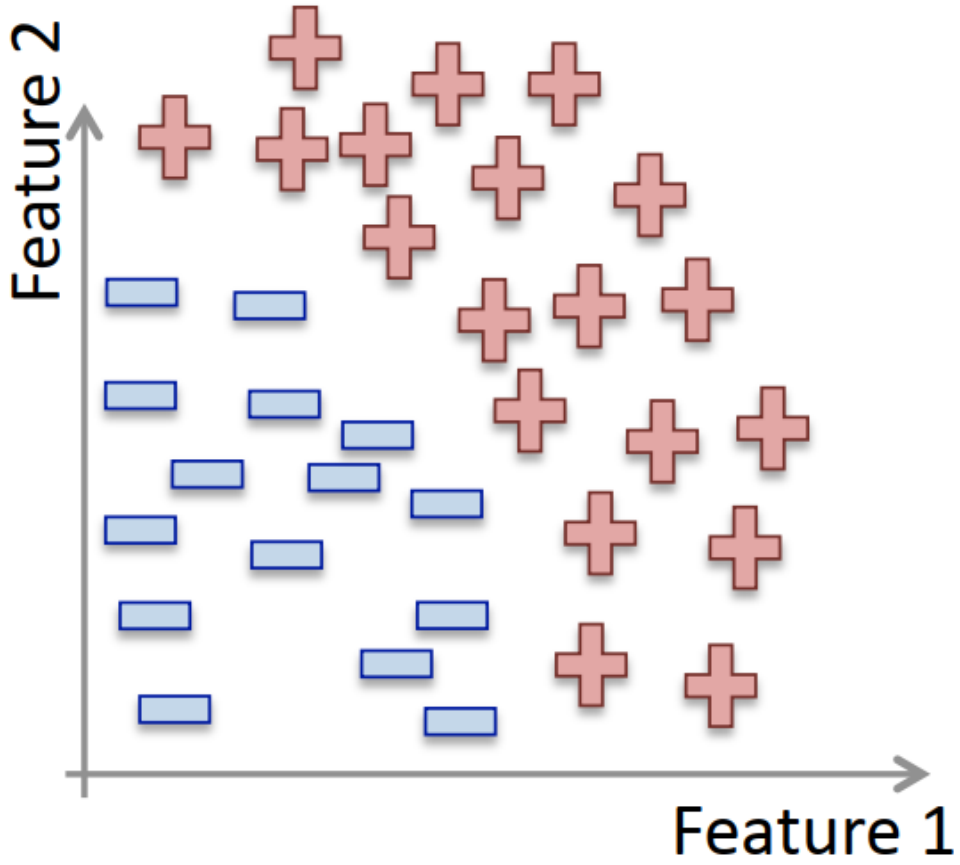
Machine Learning
Unsupervised Learning (Clustering Example)



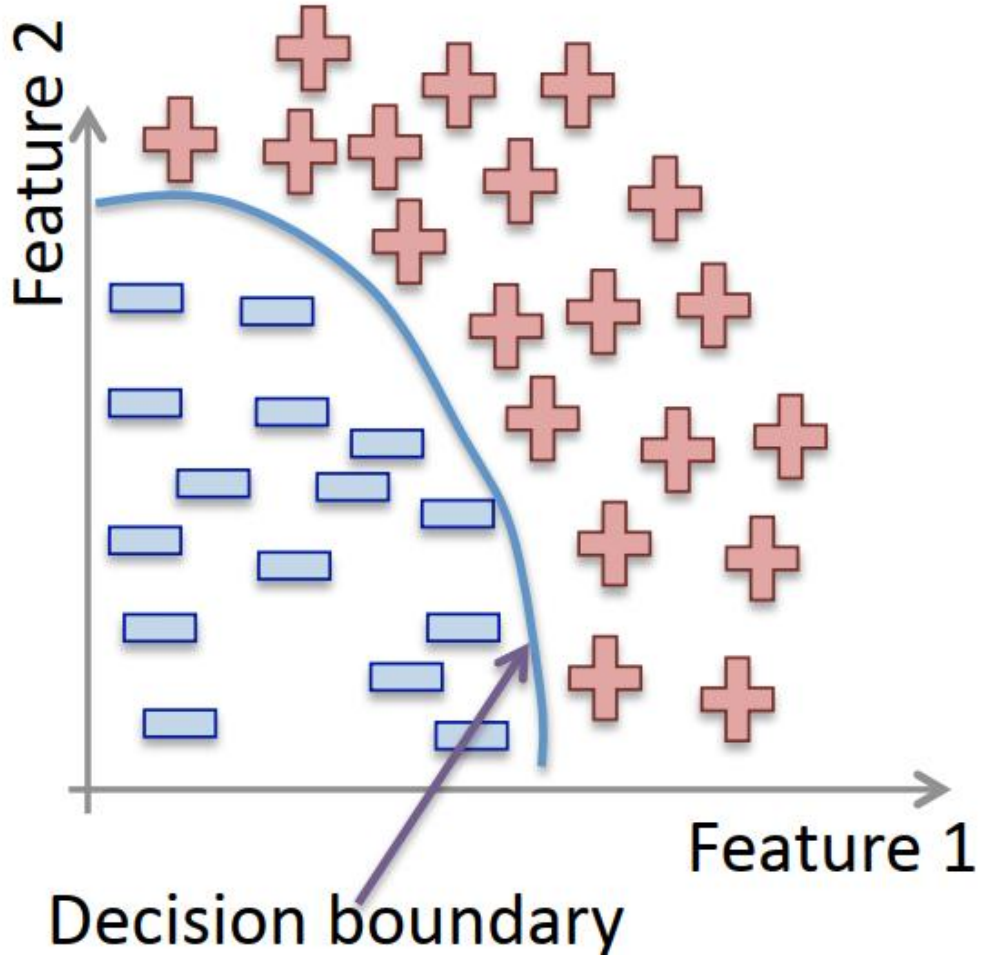
Machine Learning
Unsupervised Learning (Clustering Example)



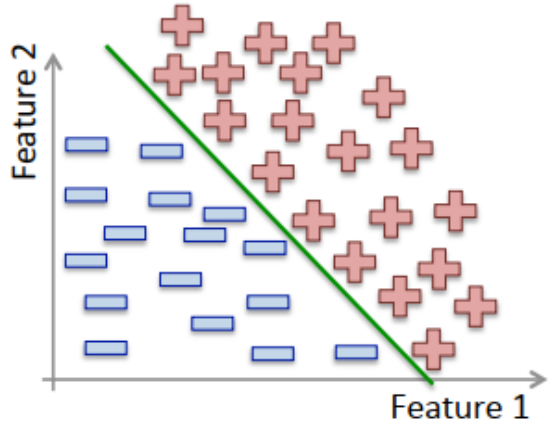
Machine Learning
Supervised Learning (Binary Classification Example)

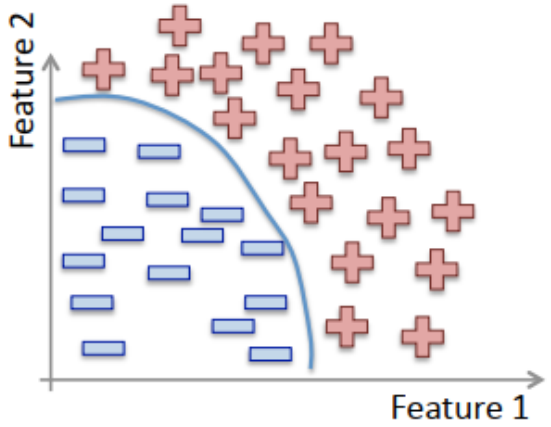


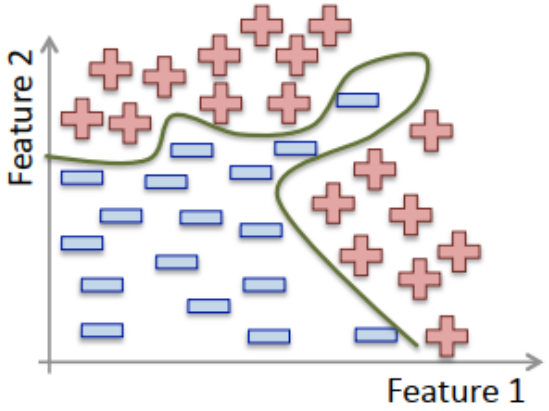
Machine Learning
Supervised Learning (Binary Classification Example)

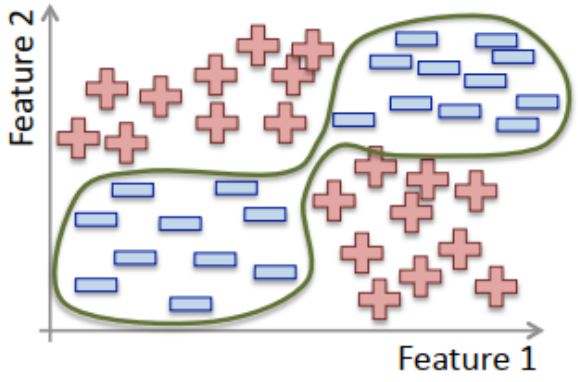


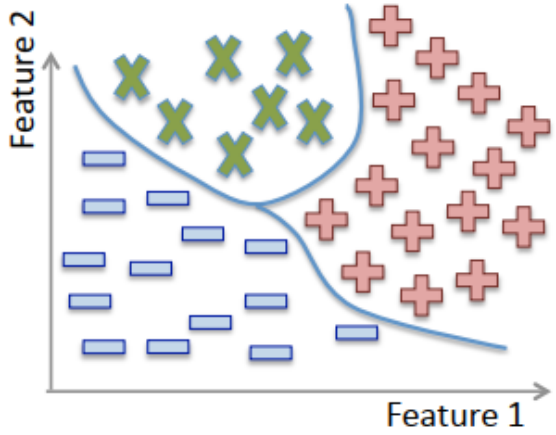
Machine Learning
Supervised Learning (Classification Example)



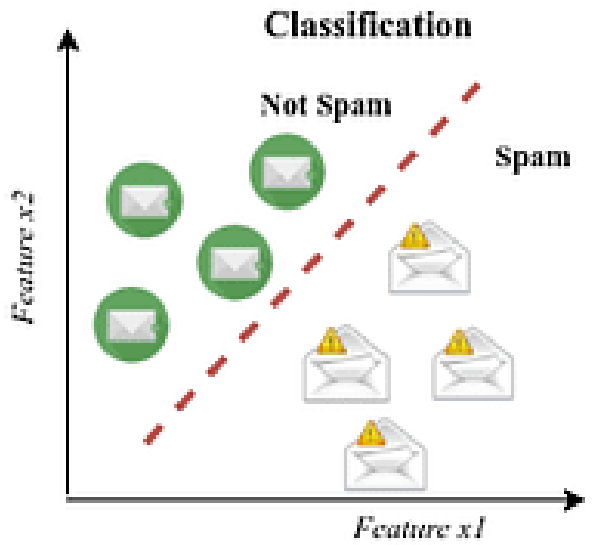


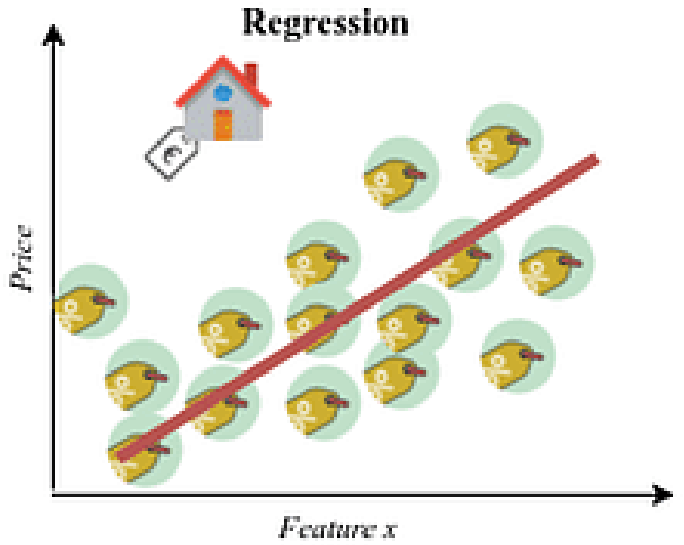


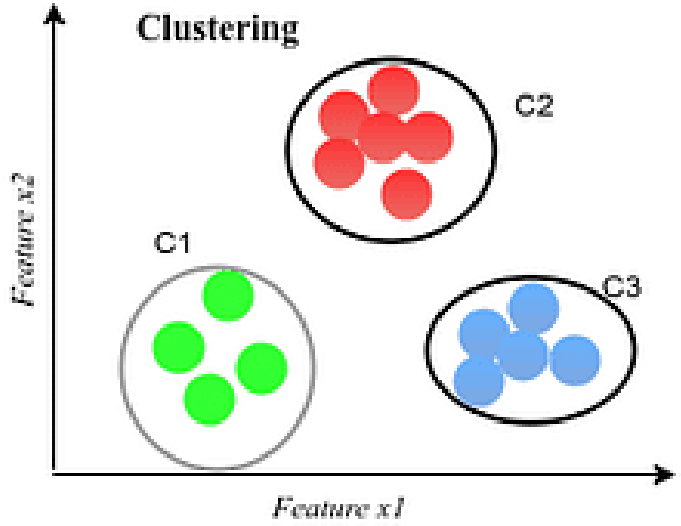




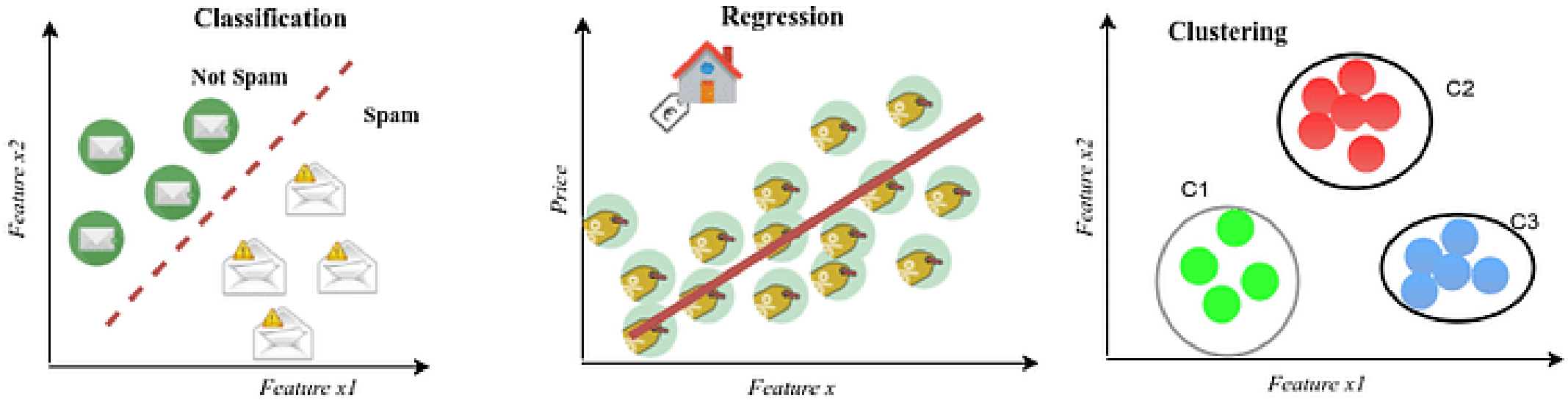
Machine Learning
Supervised and Unsupervised Learning







Machine Learning
Supervised and Unsupervised Learning



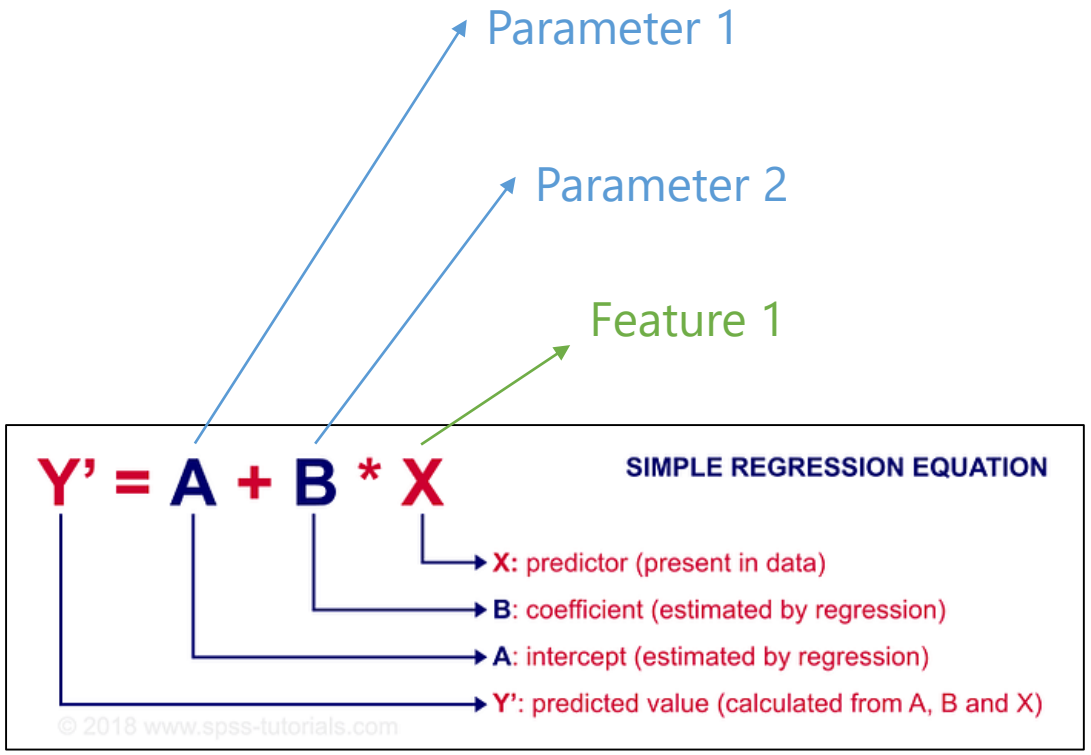
$$Y' = A + B * X$$

SIMPLE REGRESSION EQUATION

- **X:** predictor (present in data)
- **B**: coefficient (estimated by regression)
- **A**: intercept (estimated by regression)
- **Y'**: predicted value (calculated from A, B and X)

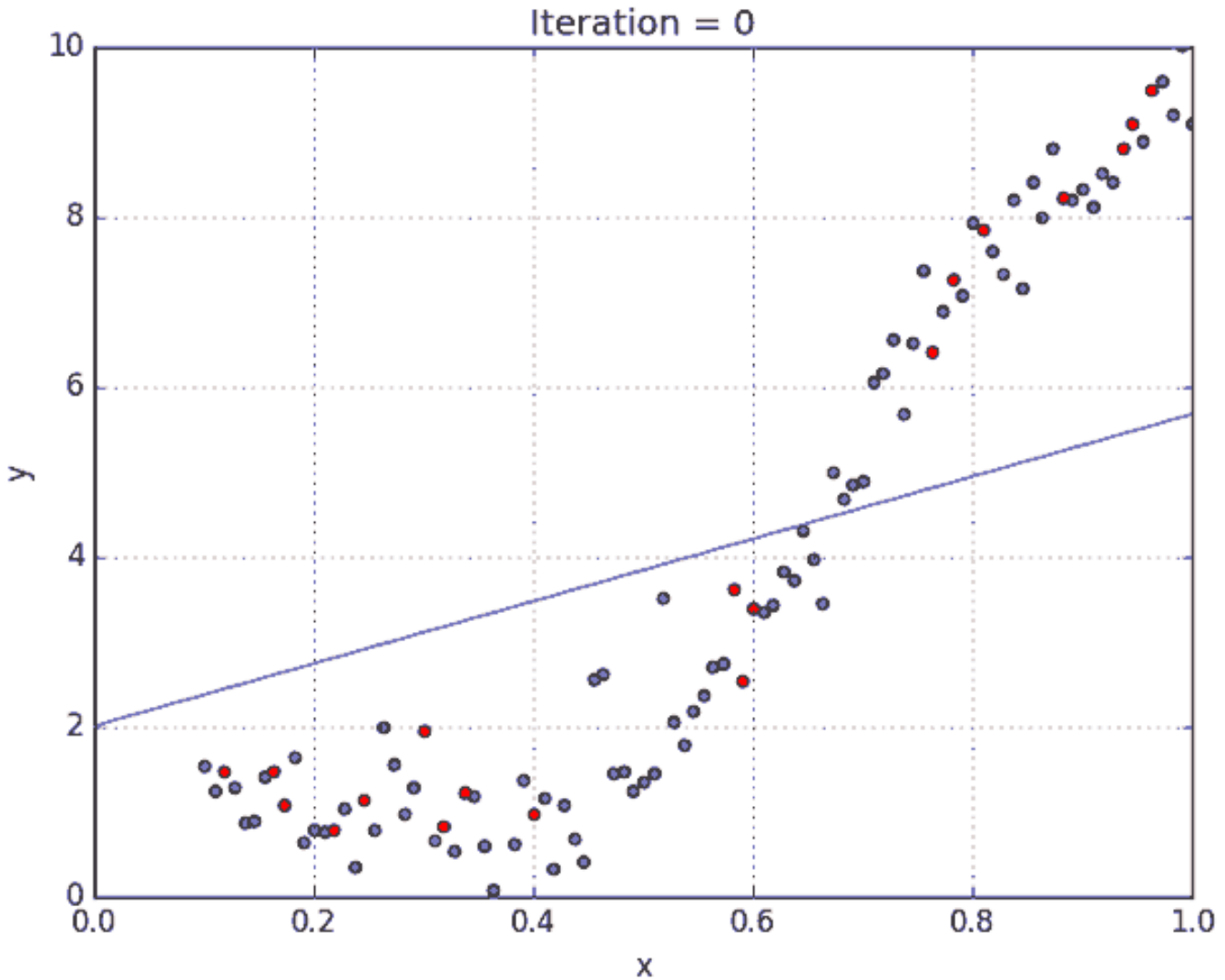
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Machine Learning
Parameters and Features

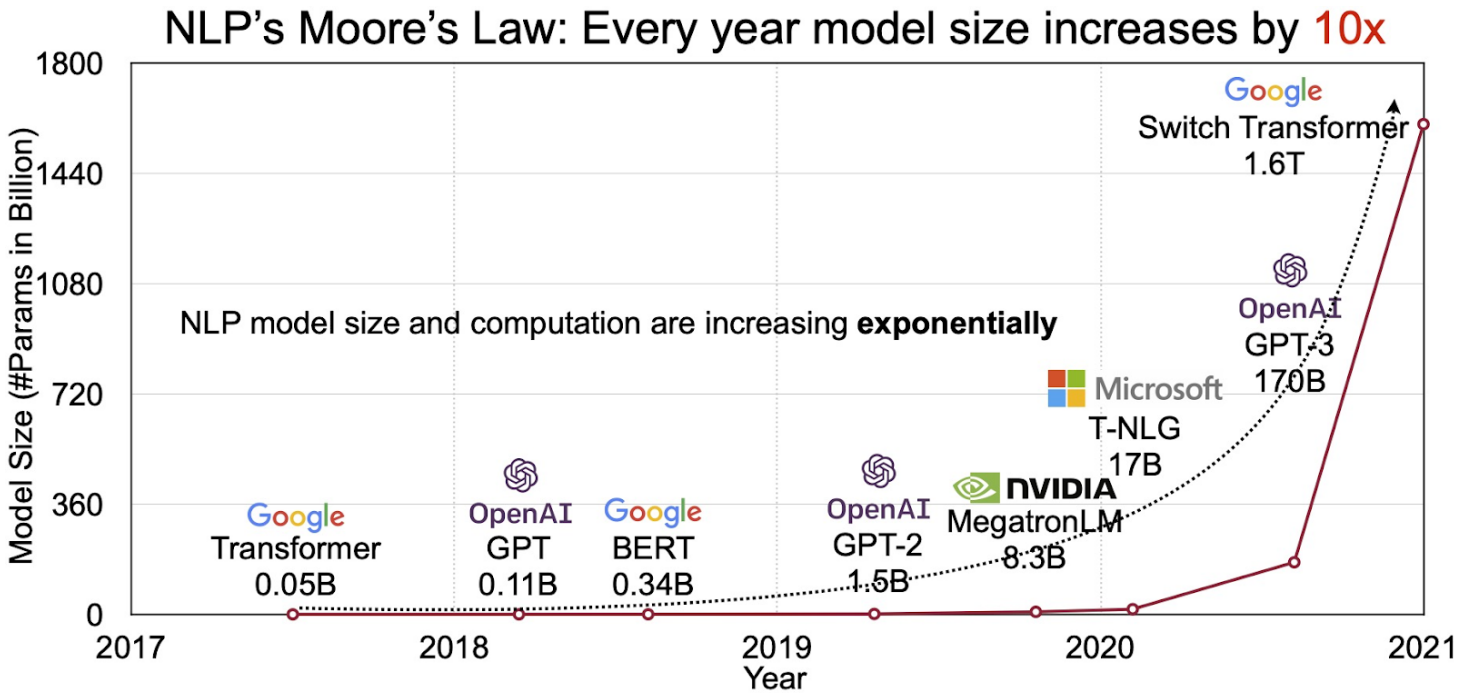


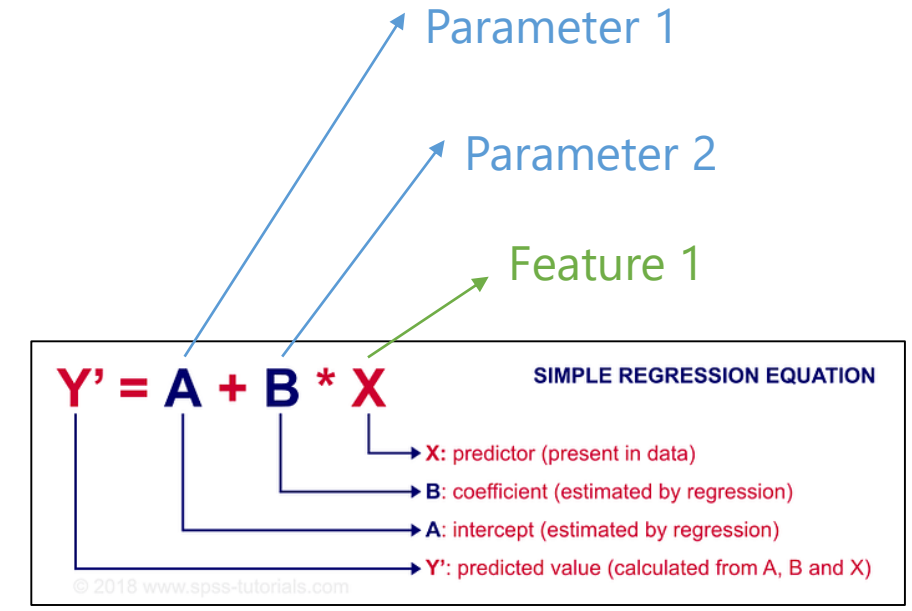
- **Features**: dependent variables; defined beforehand.
- **Parameters**: values or weights that a model uses to adjust itself during training.
- We "**train**" the **parameters** in the training phase.

Machine Learning
Parameters and Features



Machine Learning
Parameters and Features





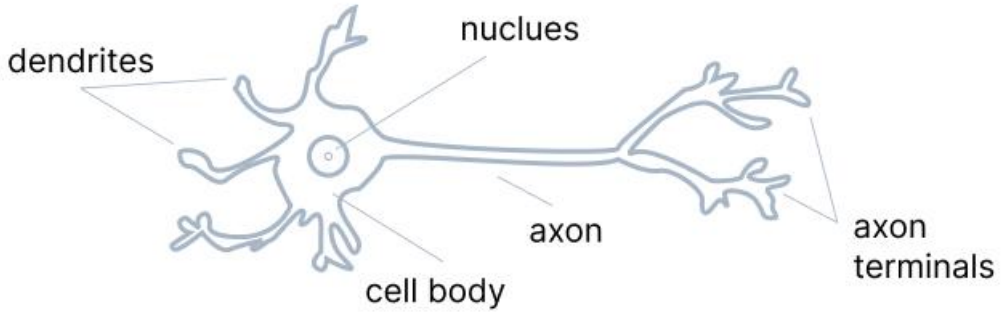
Caution: Technically not correct but it helps to get a feeling of the complexity.

Artificial Neural Networks
Introduction

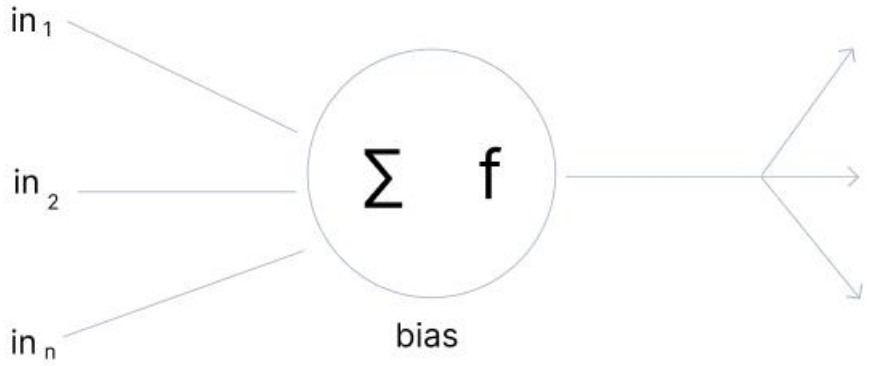
- Computational models **inspired by the human brain**:
  - Algorithms that try to mimic the brain.
  - Massively parallel, distributed system, made up of simple processing units (**neurons**)
  - **Synaptic connection** strengths among neurons are used to store the acquired knowledge.
  - Knowledge is acquired by the network from its environment through a learning process

- Artificial Neural Networks (ANN) have been widely used in various domains for:
  - Pattern recognition
  - Function approximation
  - Associative memory

Artificial Neural Networks
Neuron Definition



Biological Neuron



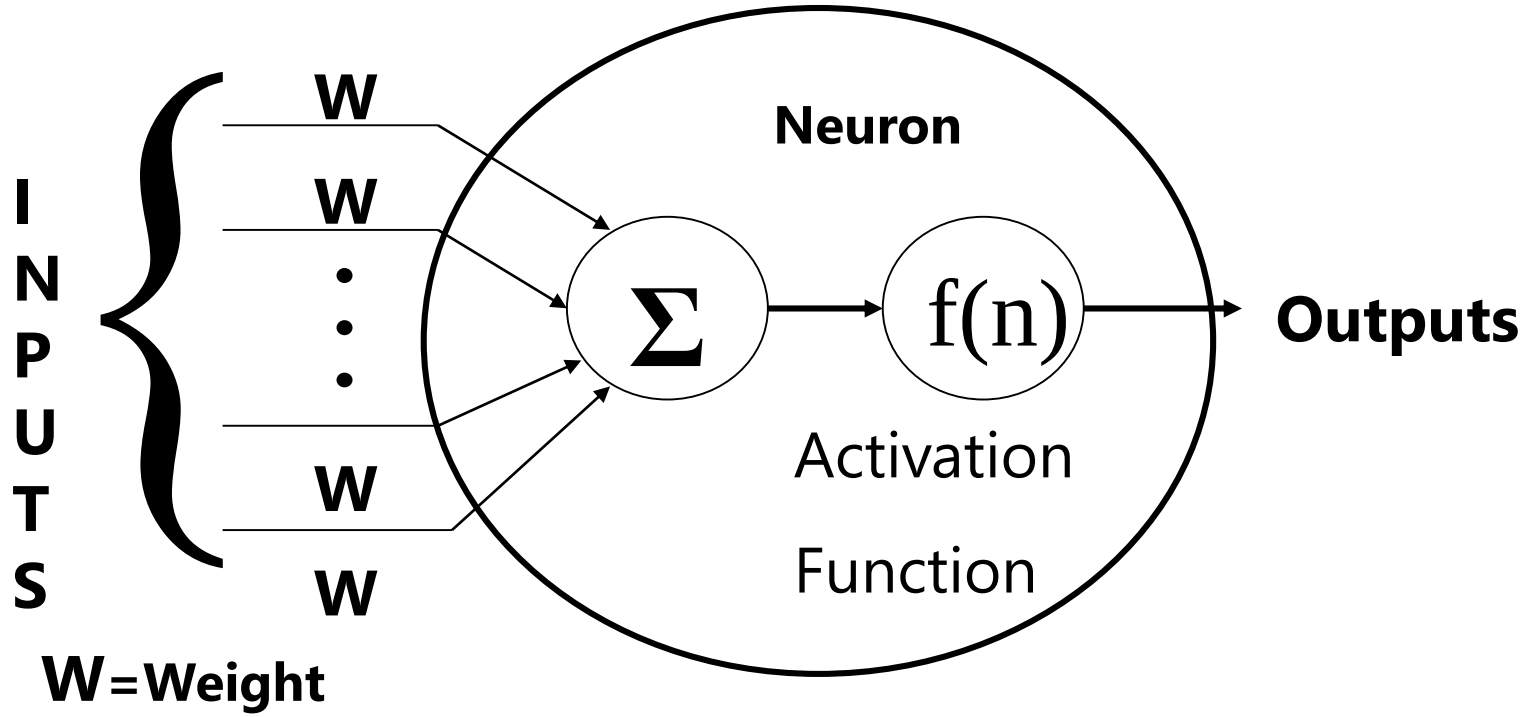
Artificial Neuron

V7 Labs

- An artificial neuron is an imitation of a human neuron

Artificial Neural Networks
Neuron Definition

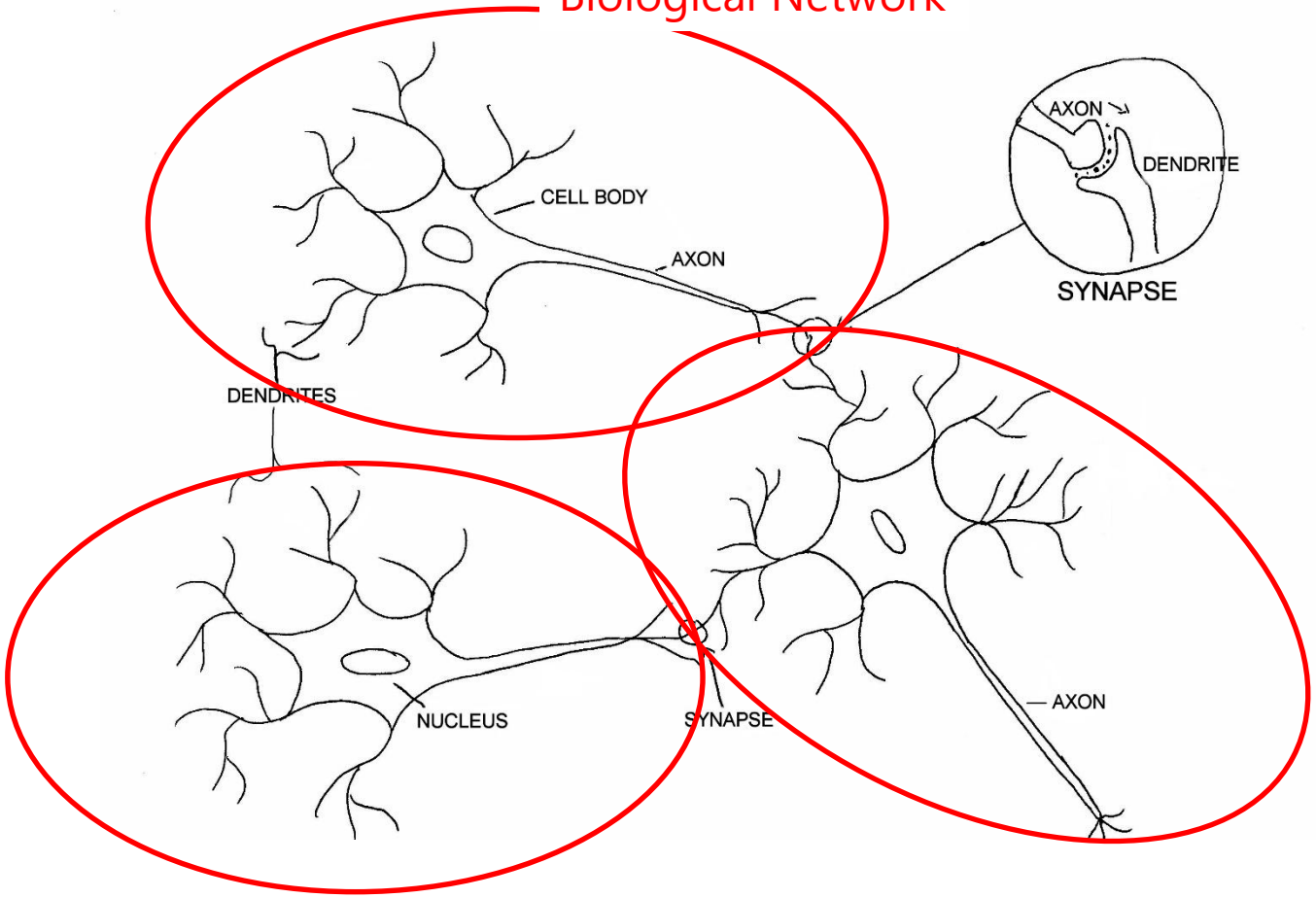
- Basic unit in a neural network is a neuron.
- It receives inputs and decides on its own if this neuron should be activated or not.
- If its activated, it will calculate and provide the result (outputs).



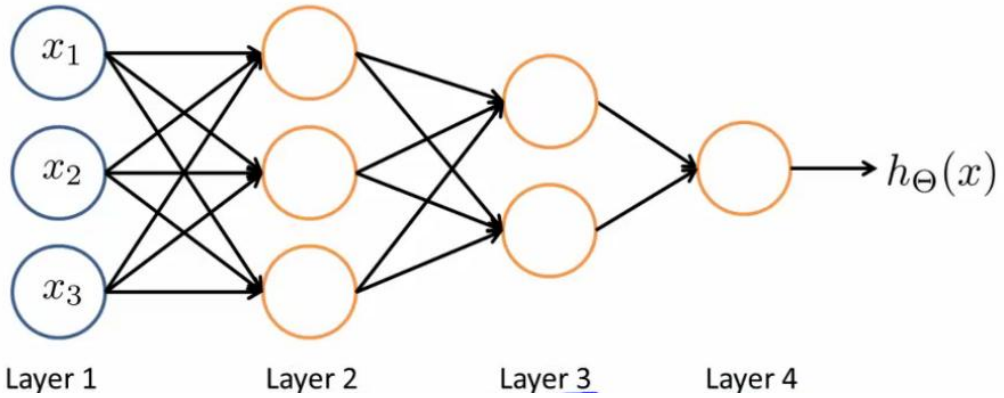
Artificial Neural Networks
Neuronal Network Definition

- Multiple neurons are connected to each other (neural network)

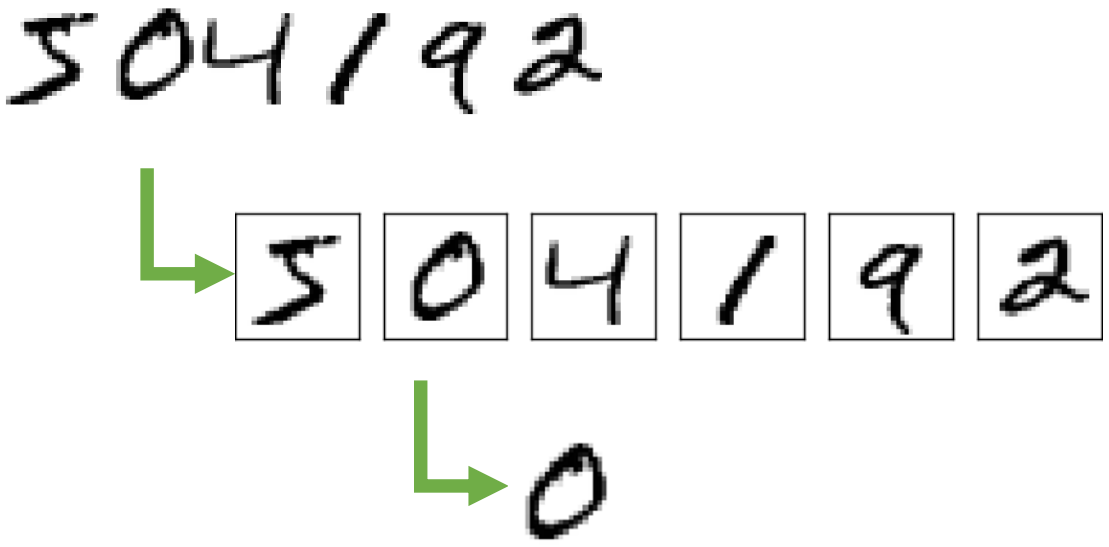
Biological Network



Artificial Network

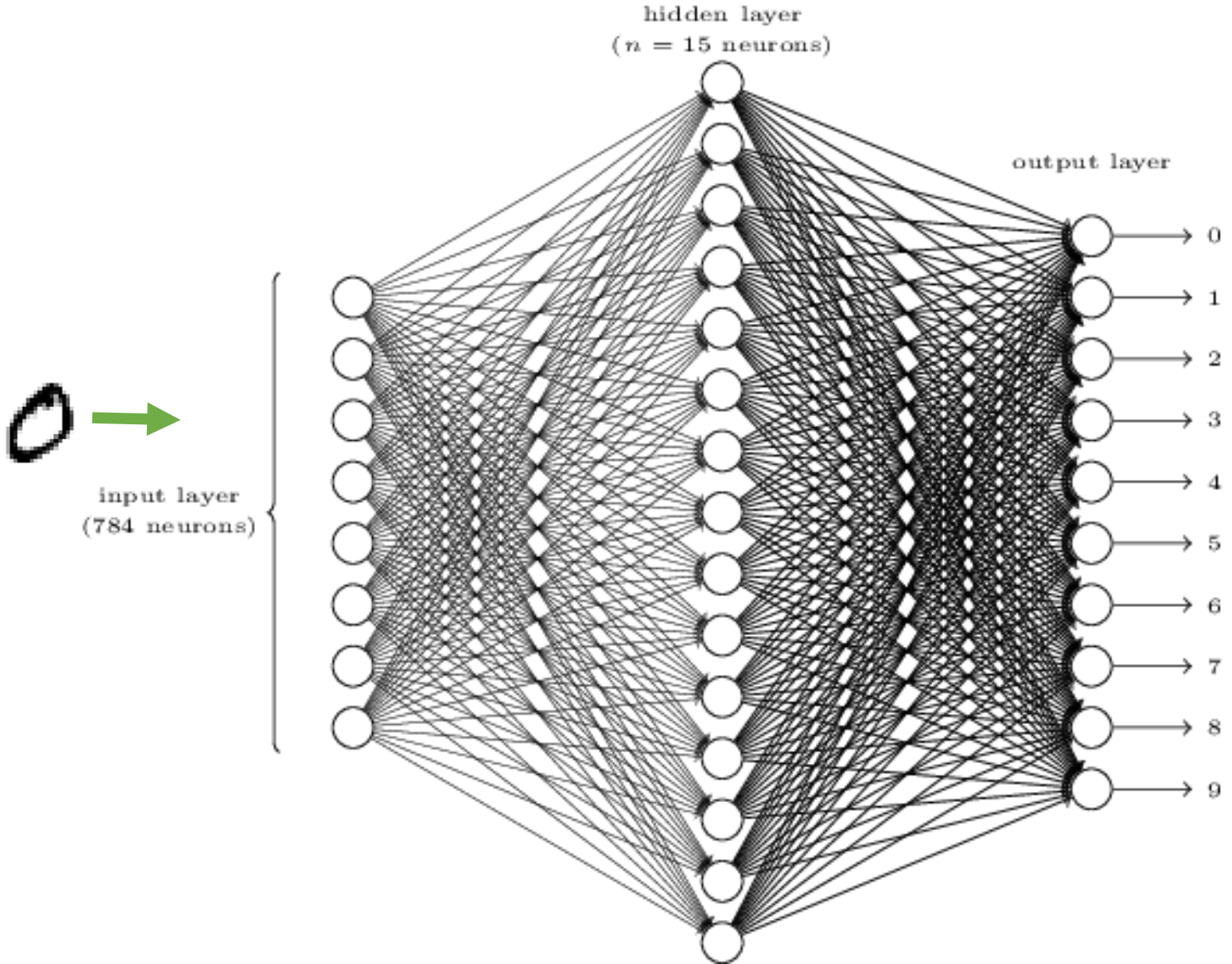


Artificial Neural Networks
Example Handwritten Number Recognition



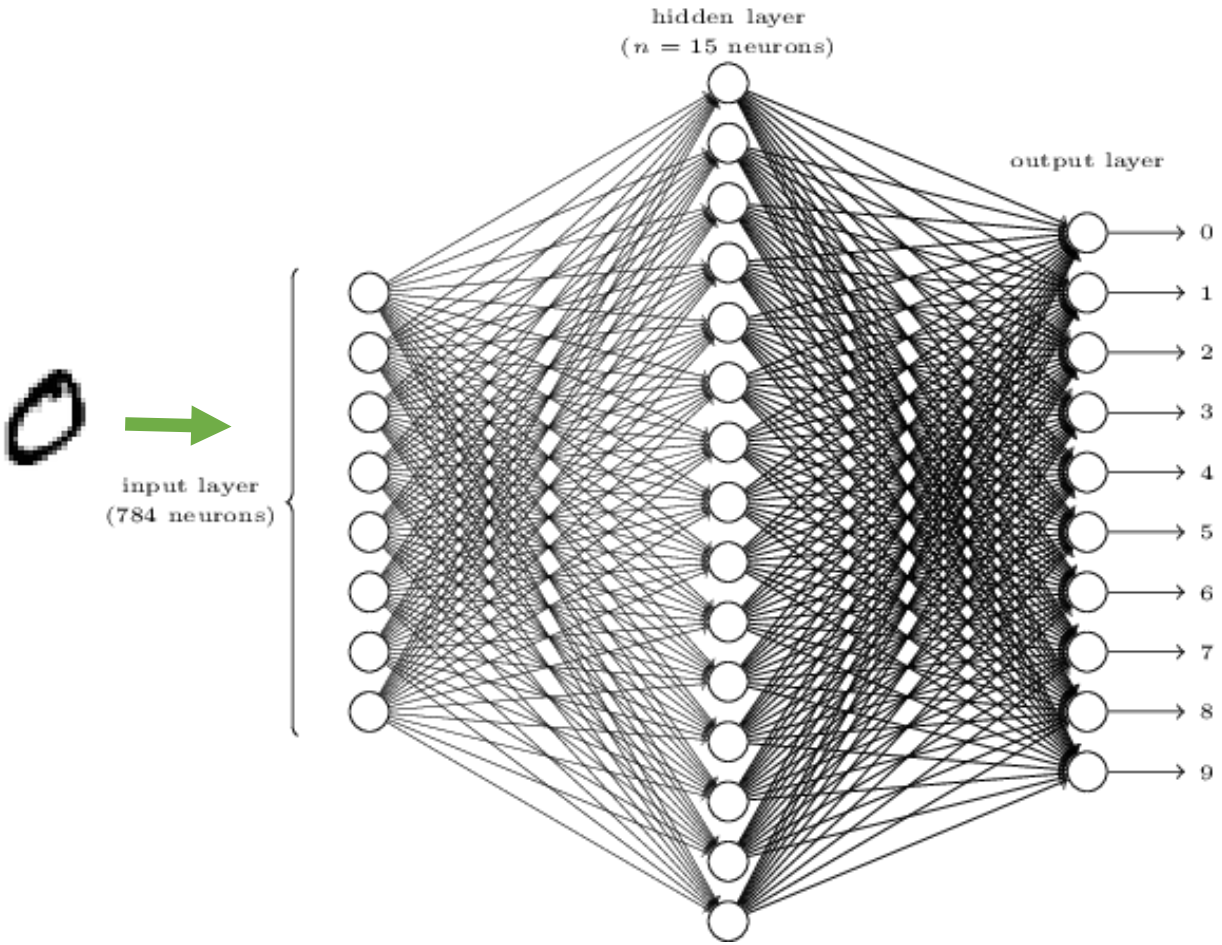
- Defining the properties:
  - We split the large number in a series of smaller numbers (0-10).
  - Image consists of 28x28=784 pixels.
  - pixels are greyscale, with a value of 0.0 (white) and 1.0 (black)

Artificial Neural Networks
Example Handwritten Number Recognition



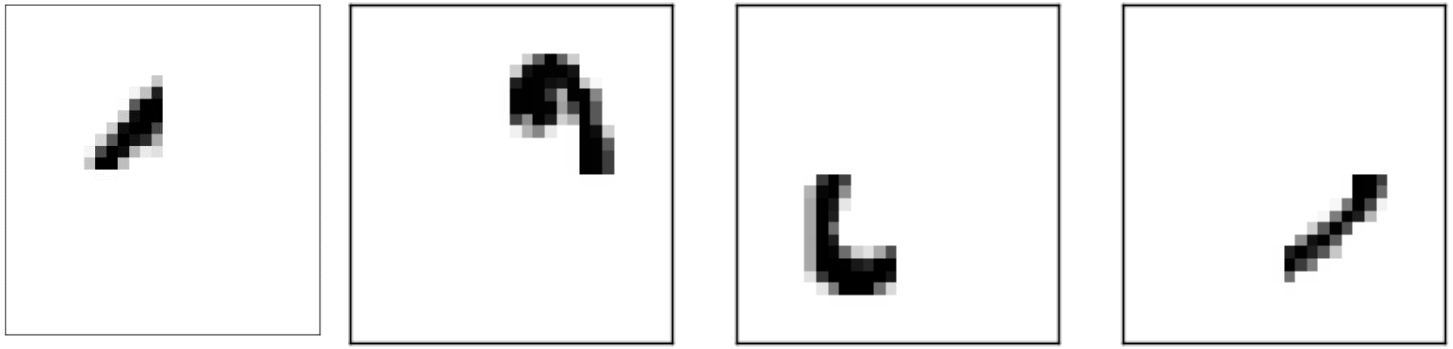
- Three **layer** neuronal net
- **Each input** is connected to each neuron.
- **Each neuron** of a layer is connected to any neuron of the next layer.

Artificial Neural Networks
Example Handwritten Number Recognition



- **Hidden layers** are the intermediary stages between input and output.
- The hidden layer represent the **learning.**
- **Each neuron in the layer** "solves a part of the problem".

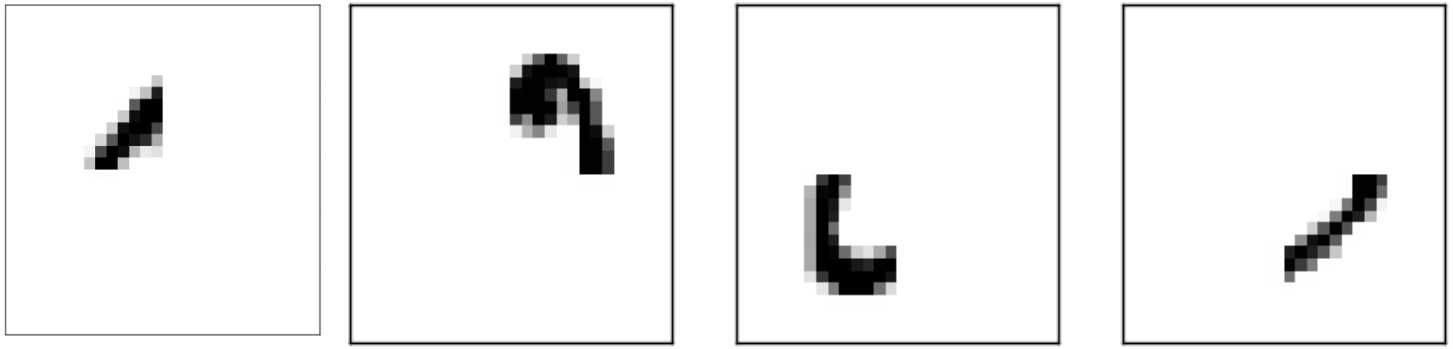
Artificial Neural Networks
Example Handwritten Number Recognition



- The four images represents the first four neurons of the hidden layer.
- Each neuron consists of a part of the solution.
- When a pixel arrives, it validates if the pixel matches the neuron (black or white).
- The first four neurons together represent the „0“ handwritten number.

Caution: For illustration purpose only.

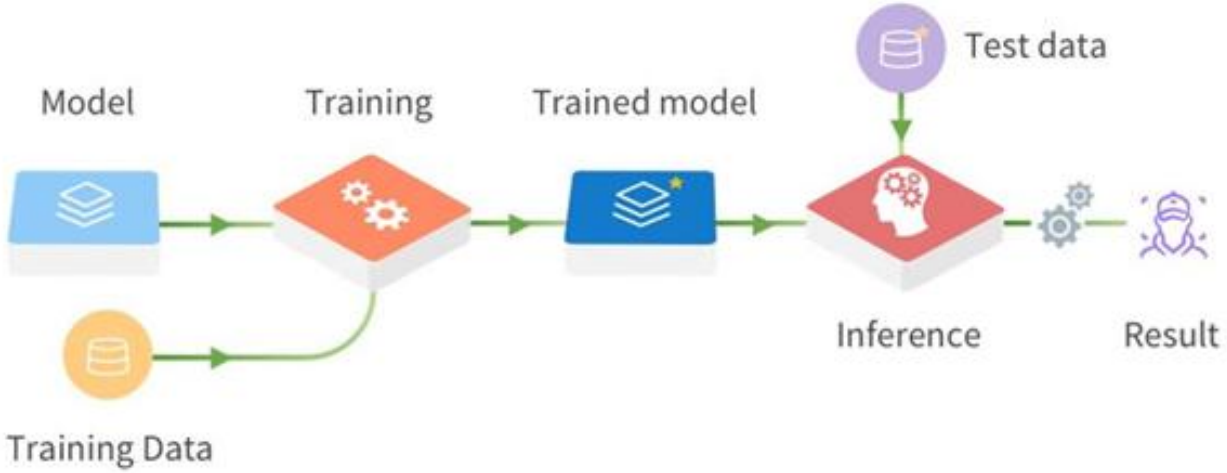
Artificial Neural Networks
Example Handwritten Number Recognition



- The first four neurons together represent the „0“ handwritten number.
- If this four neurons are activated, the output layer also activates the 0-neuron.

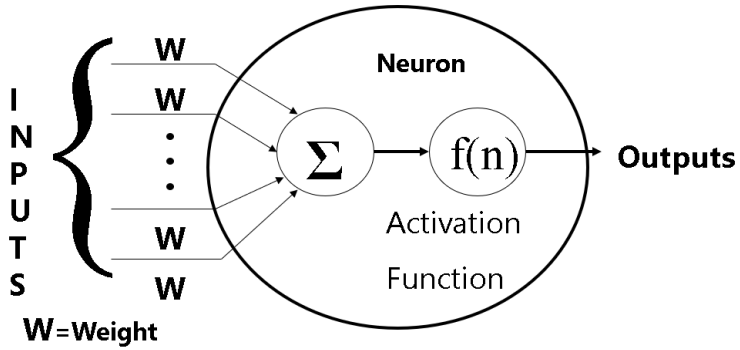
- In real-world its not that simple.
- We have different handwriting styles which all have to take into account in this neurons to recognize the “0”.

Artificial Neural Networks
Training and Testing



- Training process at a high level is similar to machine learning.

- What we „**train**“ are the **weight-parameters** of all neurons.



Artificial Neural Networks
Training and Learning and Applications

- Real-world applications consists of >100 hidden layers.
- Each layer consists of >10000 neurons.

- The number of inputs are typically >10000.
- The number of outputs depends on the use case.

Deep Learning
Introduction

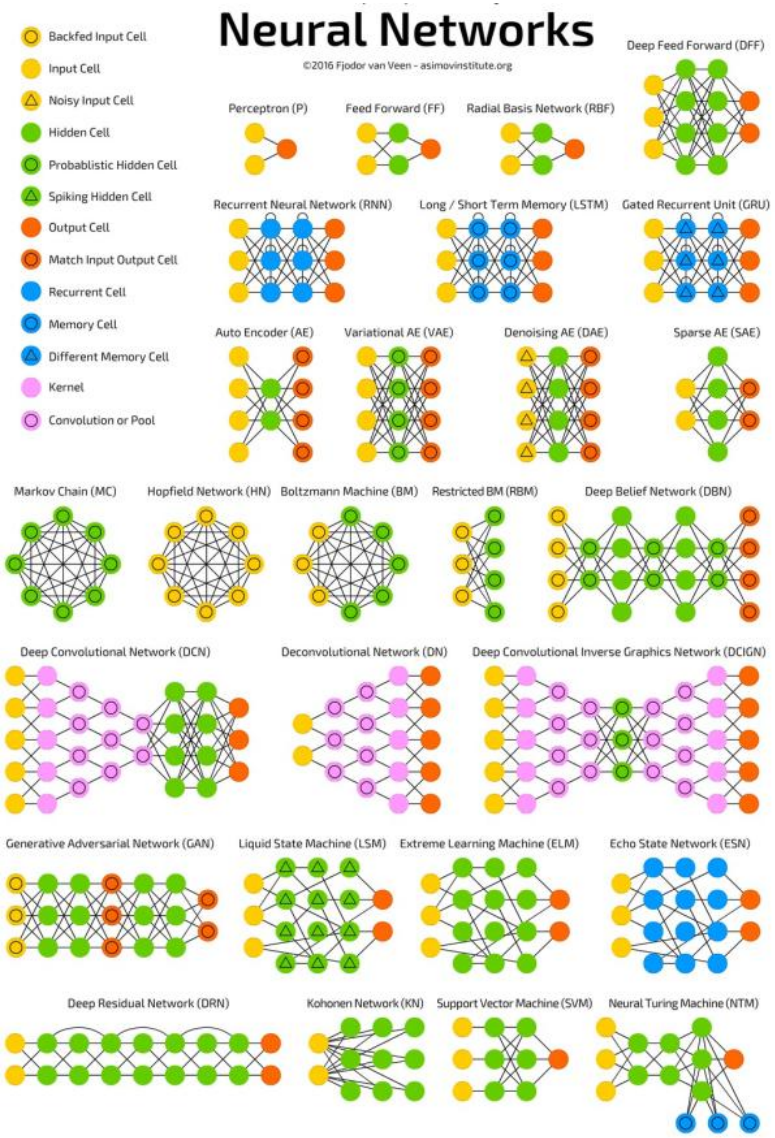
- A deep neural network is a neural network with three or more layers.
- In practice, most DNNs have many more layers.
- DNNs are trained on large amounts of data to identify and classify phenomena, recognize patterns and relationships, evaluate possibilities, and make predictions and decisions.

- While a single-layer neural network can make useful, approximate predictions and decisions, the additional layers in a deep neural network help refine and optimize those outcomes for greater accuracy.

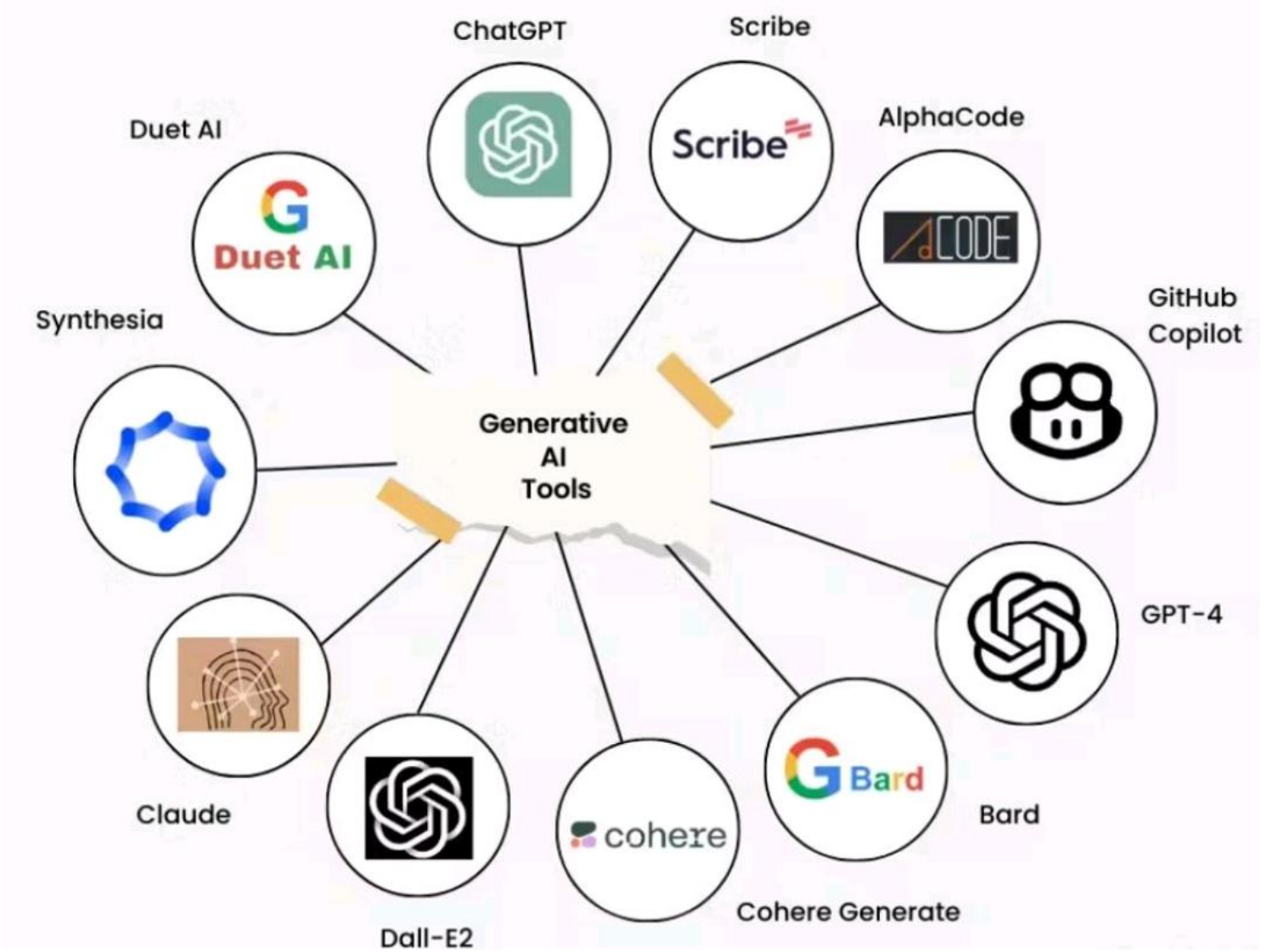
Deep Learning
Introduction

	Deep learning systems	Simple neural networks
Architecture	Consists of several hidden layers arranged for convolution or recurrence.	Neural networks consist of an input, hidden, and output layer. They mimic the human brain in structure.
Complexity	Depending on its function, a deep learning network is highly complicated and has structures like long short-term memory (LSTM) and autoencoders.	Neutral networks are less complicated, as they have only a few layers.
Performance	A deep learning algorithm can solve complex issues across large data volumes.	Neural networks perform well when solving simple problems.
Training	It costs a lot of money and resources to train a deep learning algorithm.	The simplicity of a neural network means it costs less to train.

Deep Learning and Neural Networks
Architecture Overview



Large Language Models
Introduction

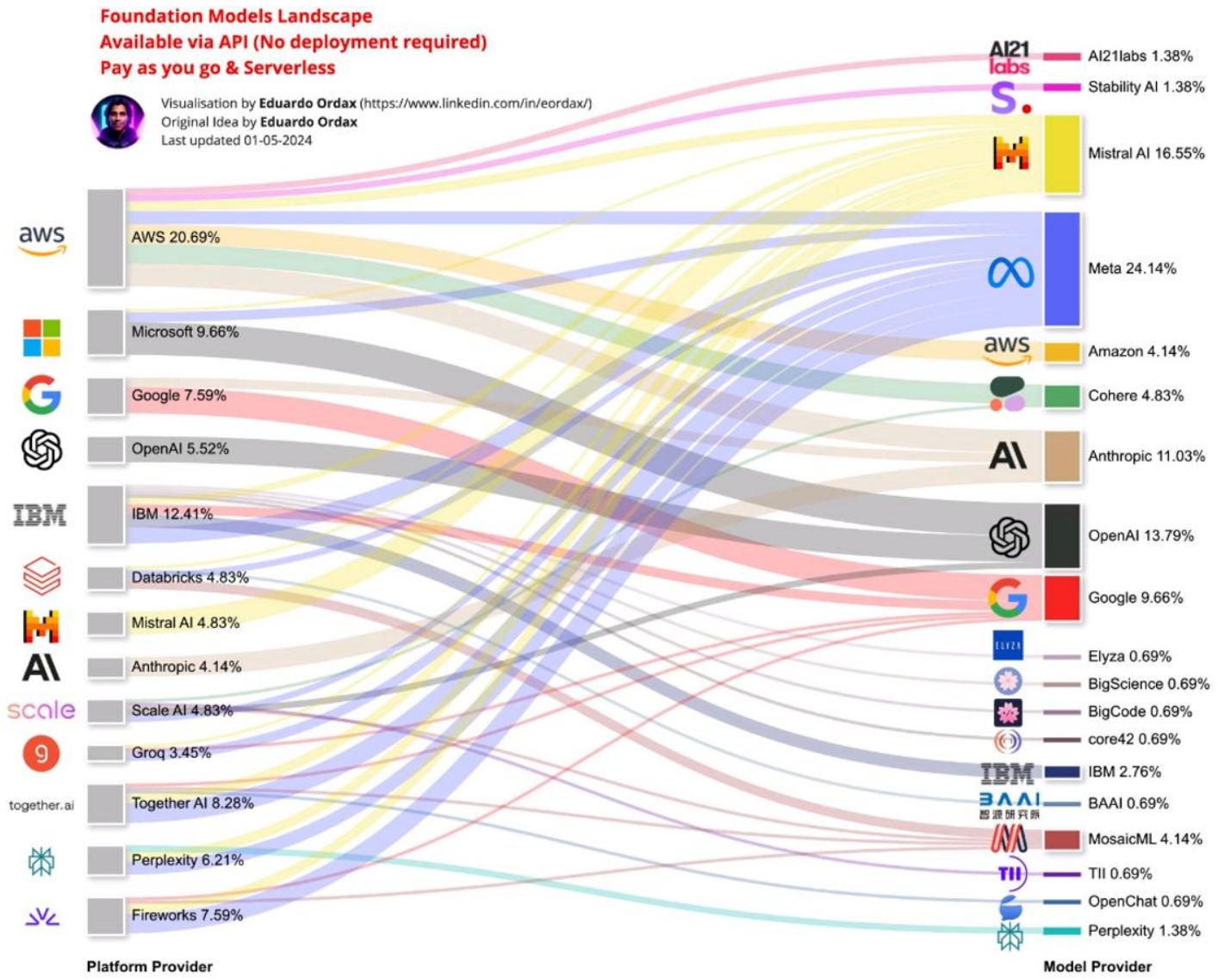


Large Language Models
Introduction

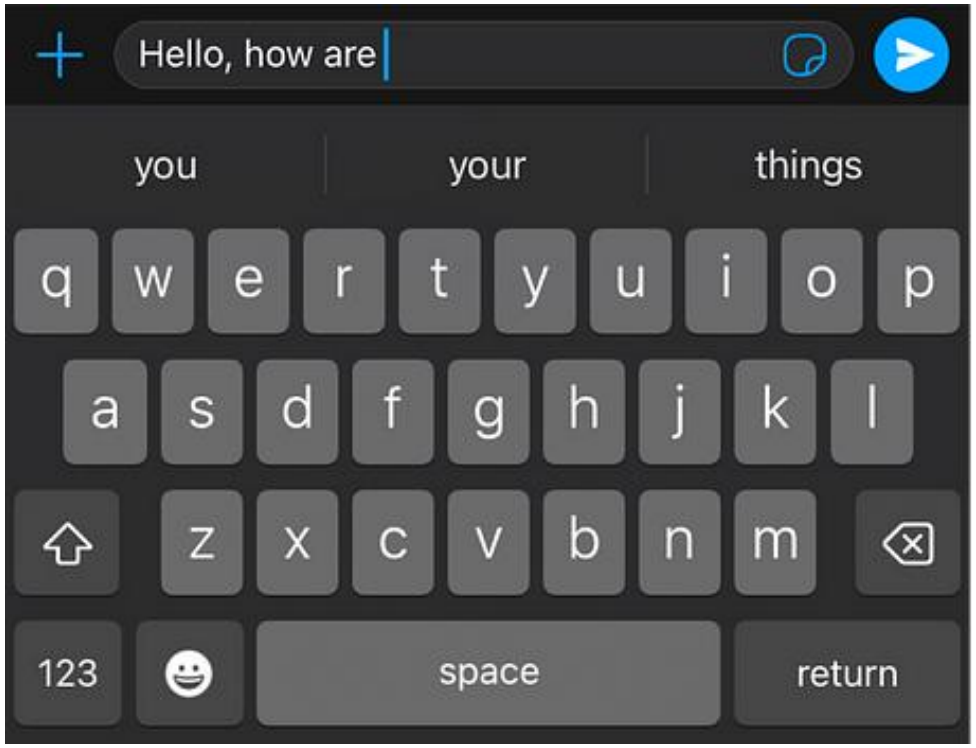
- A type of deep learning model designed to process and understand natural language data (NLP).
- Called „*large*“ since they are trained on large amount of data.

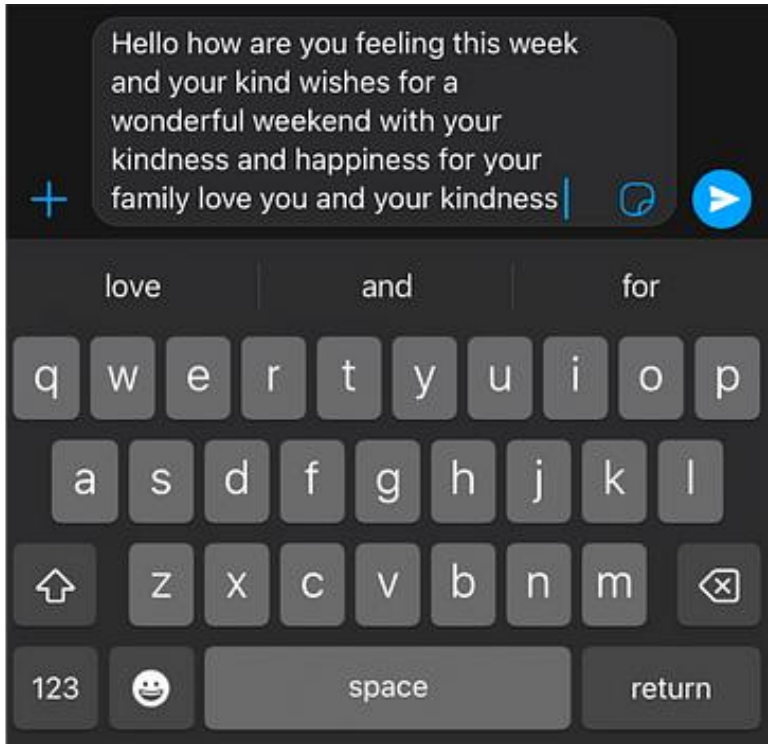
- They are built on the **Transformer** neural network architecture.
- Introduced in 2017 by Google.
- Becomes popular in 2023 since ChatGPT was released.

Large Language Models
Introduction



Large Language Models (LLMs)
Introduction





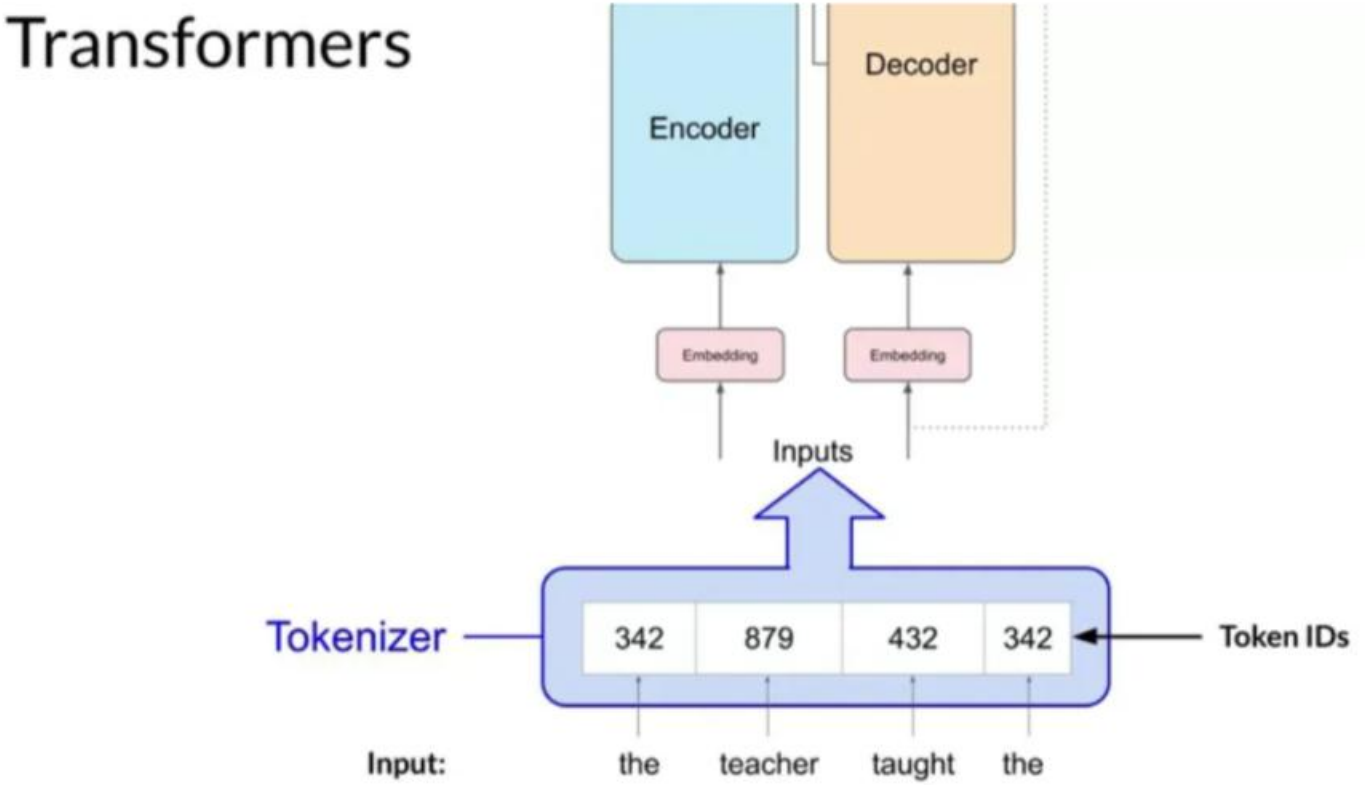
- Existing approaches based on Recurrent Neural Network (RNN) DL architecture.
- RNNs only incorporate a few words and guess the next possible word (without context)

Large Language Models
Transformers



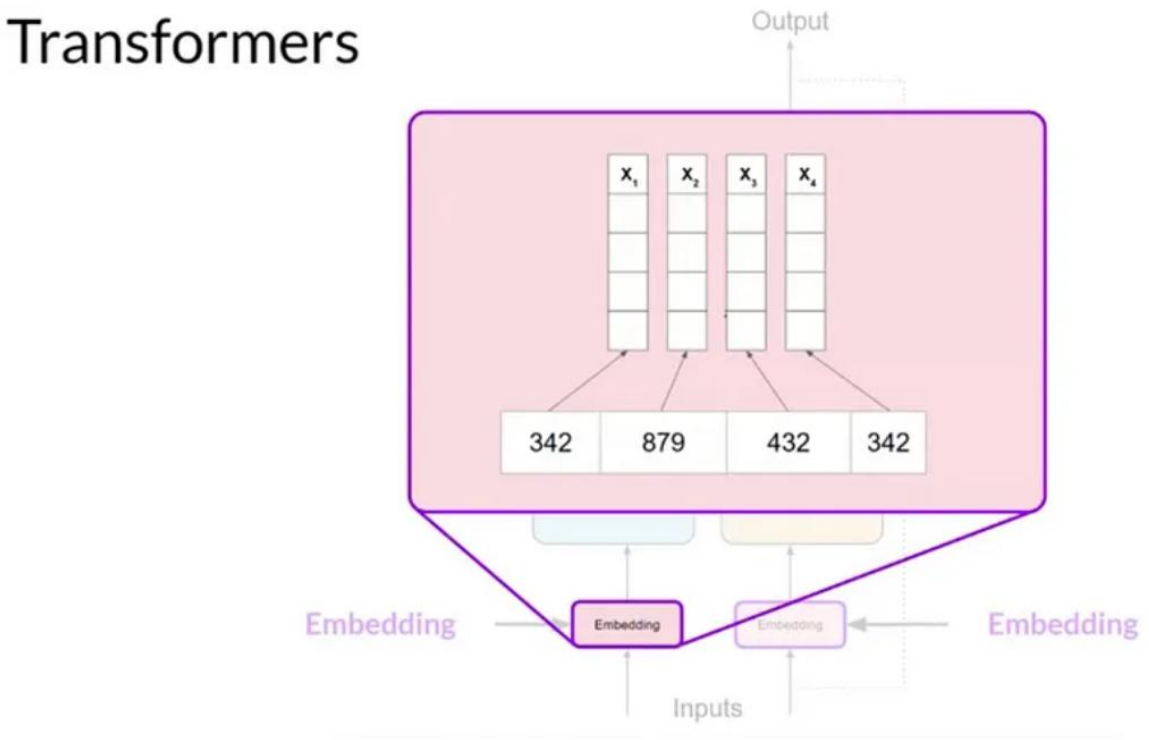
- Enable models to handle long-range dependencies in text.
- Rich context could be determined and considered.
- Not only useful for NLP
  - **Machine translation**: automatic speech translation, code translation, …
  - **DNA sequence analysis**: predict the effects of genetic mutations, …
  - **Protein structure analysis**: model long chains of amino acids that fold into complex protein structures

Large Language Models
Transformer Architecture - Tokenizing



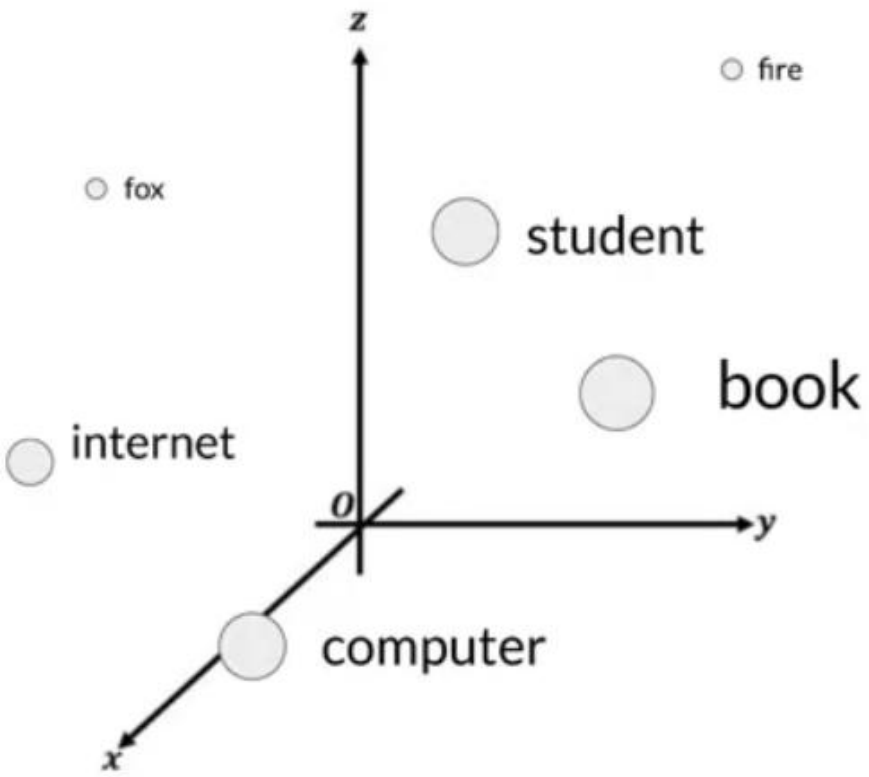
- Transform the words into numbers (LLMs are statistical models: numbers only).
- Tokens show IDs which represents the word into the world universe of the previously trained model.

Large Language Models
Transformer Architecture - Embedding



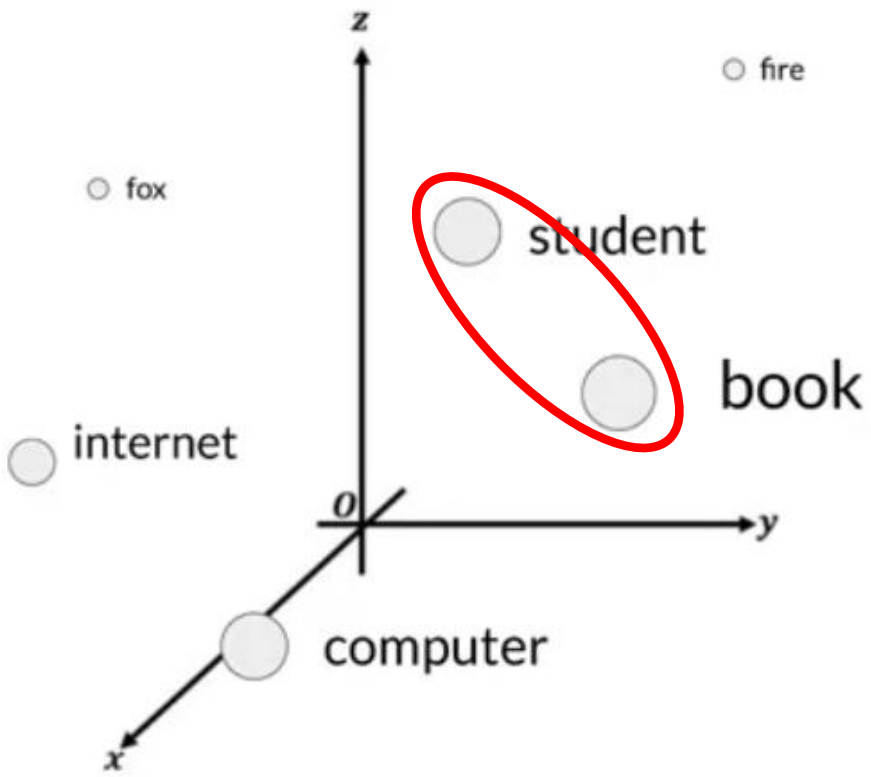
- Since the input is represented as numbers, you can pass it to the embedding layer.
- Each word has been matched to a token ID, and each token is mapped into a vector.

Large Language Models
Transformer Architecture - Embedding



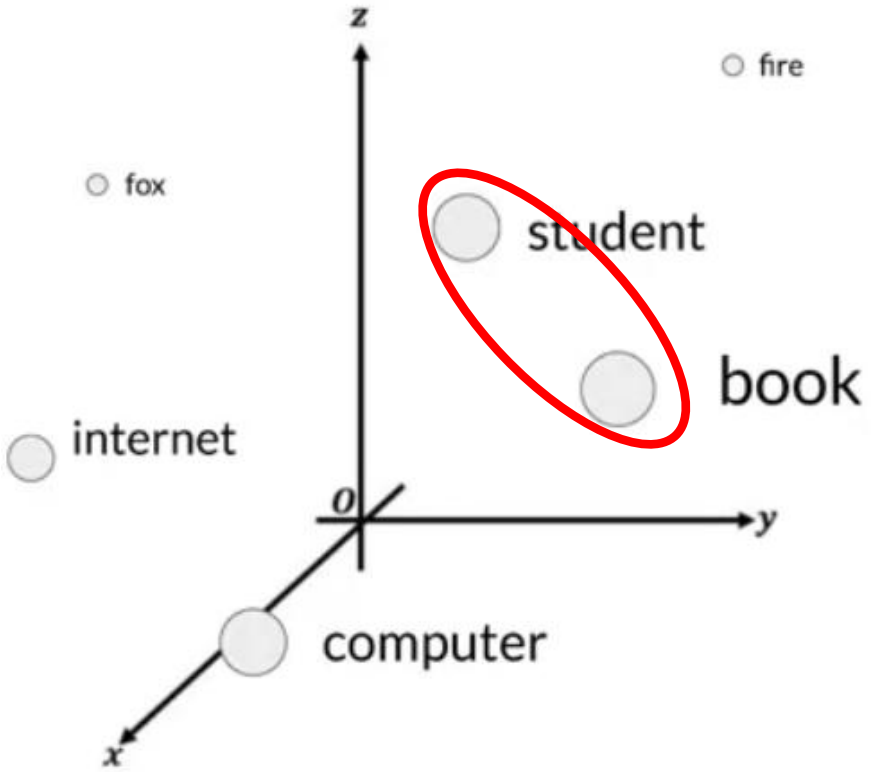
- Imagine a vector size of three, you could plot the words into a three-dimensional space and see the relationships between those words.

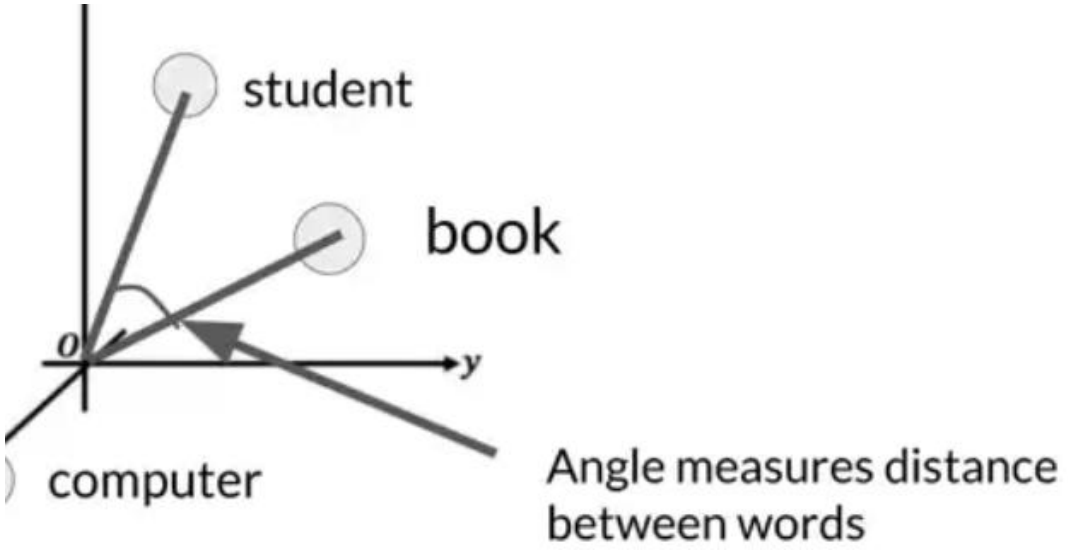
Large Language Models
Transformer Architecture - Embedding



- Words which are “close” together are related to each other.
- How can we measure the “distance”?

Large Language Models
Transformer Architecture - Embedding





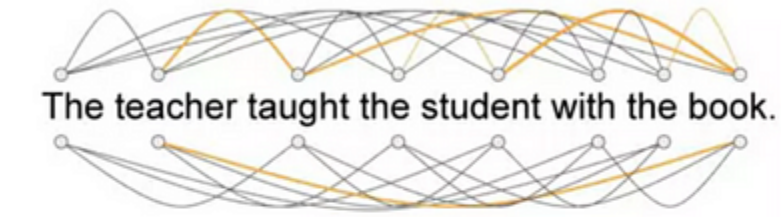
- Words which are “close” together are related to each other.
- How can we measure the “distance”?

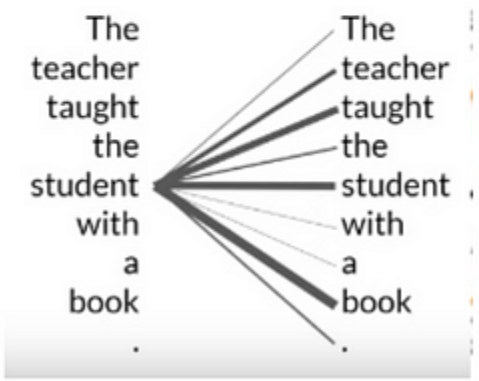
Large Language Models
Transformer Architecture – LLM Training

The power of the transformer architecture lies in its ability to learn the relevance and context of all of the words in a sentence.to every other word in a sentence.

apply attention weights to those relationships so that the model learns the relevance of each word to each other words no matter where they are in the input

These attention weights are learned during LLM training

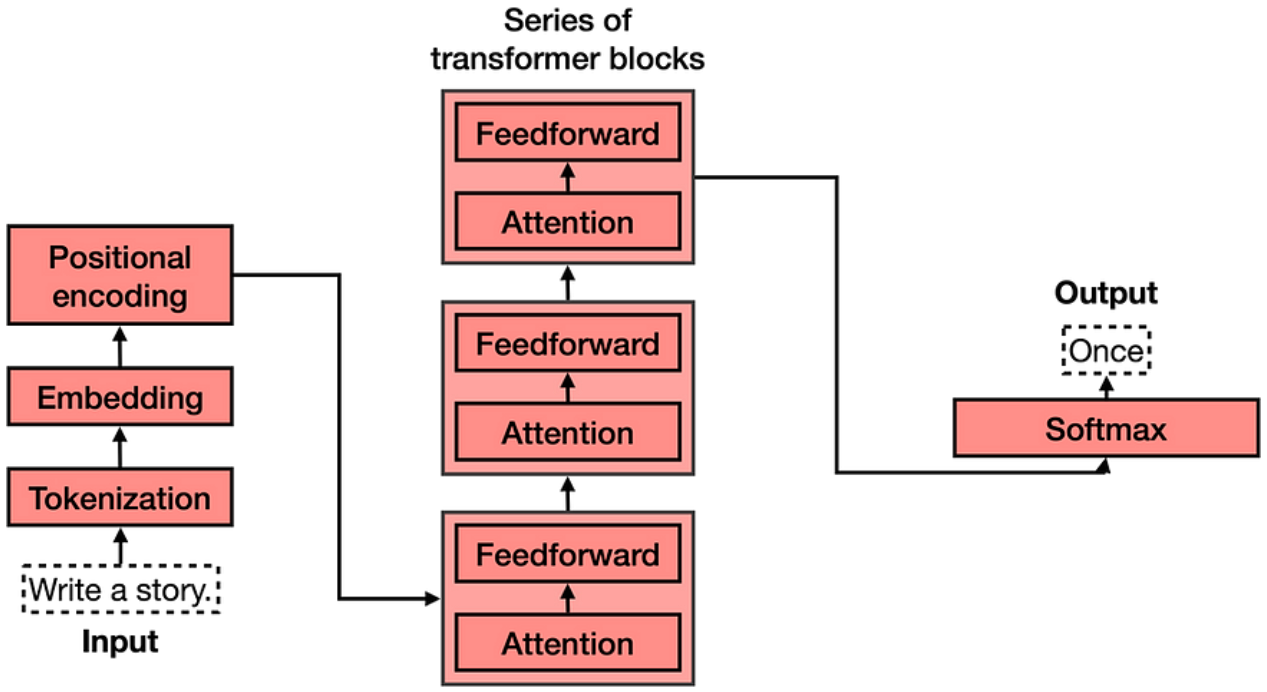




This diagram is called an attention map and can be useful to illustrate the attention weights between each word and every other word.

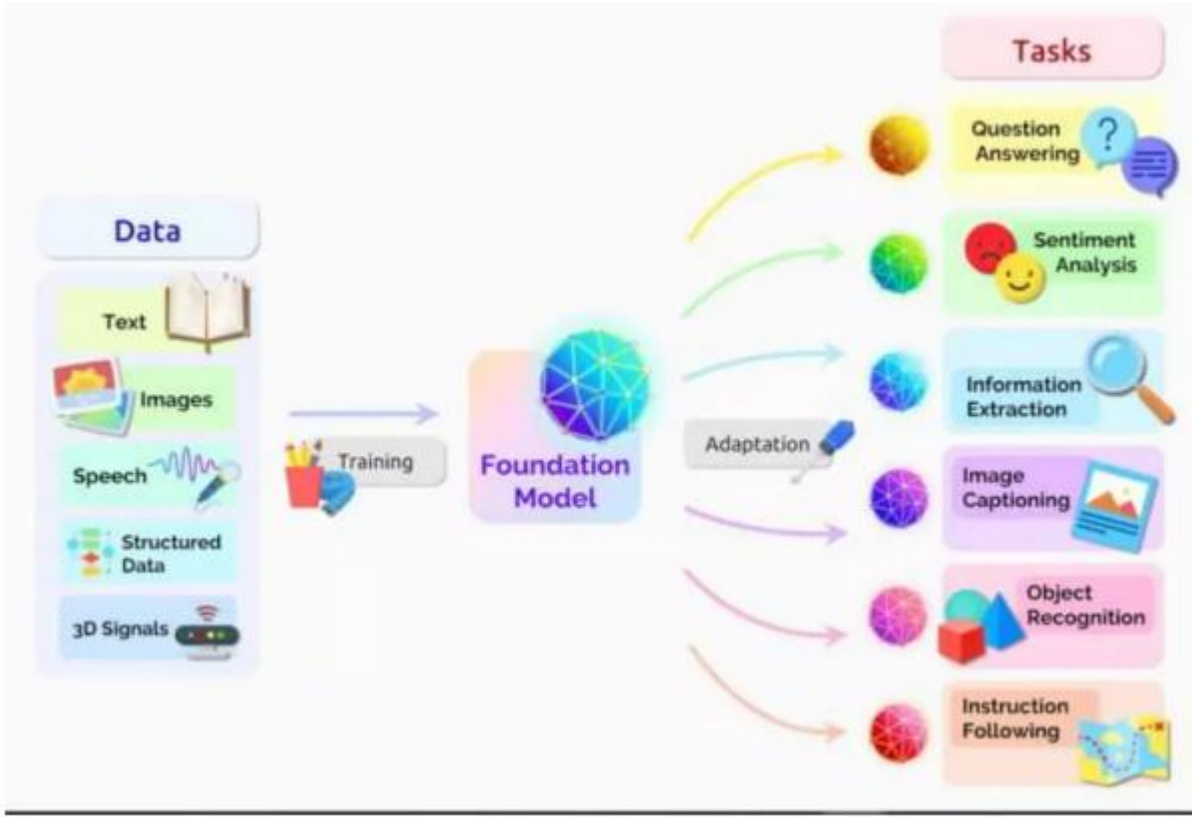
the ability to learn attention in this way across the whole input significantly approves the model's ability to encode language

Large Language Models
Transformer Architecture



- Left side as discussed
- The **Transformer** modifies the sentences in order to derive the context.
  - change words, exchange words, move words, …
- The **Softmax** function determines the „ meaningfulness“ of all transformer proposals.
- This video explains Transformers very well.

Large Language Models
Transformer Architecture

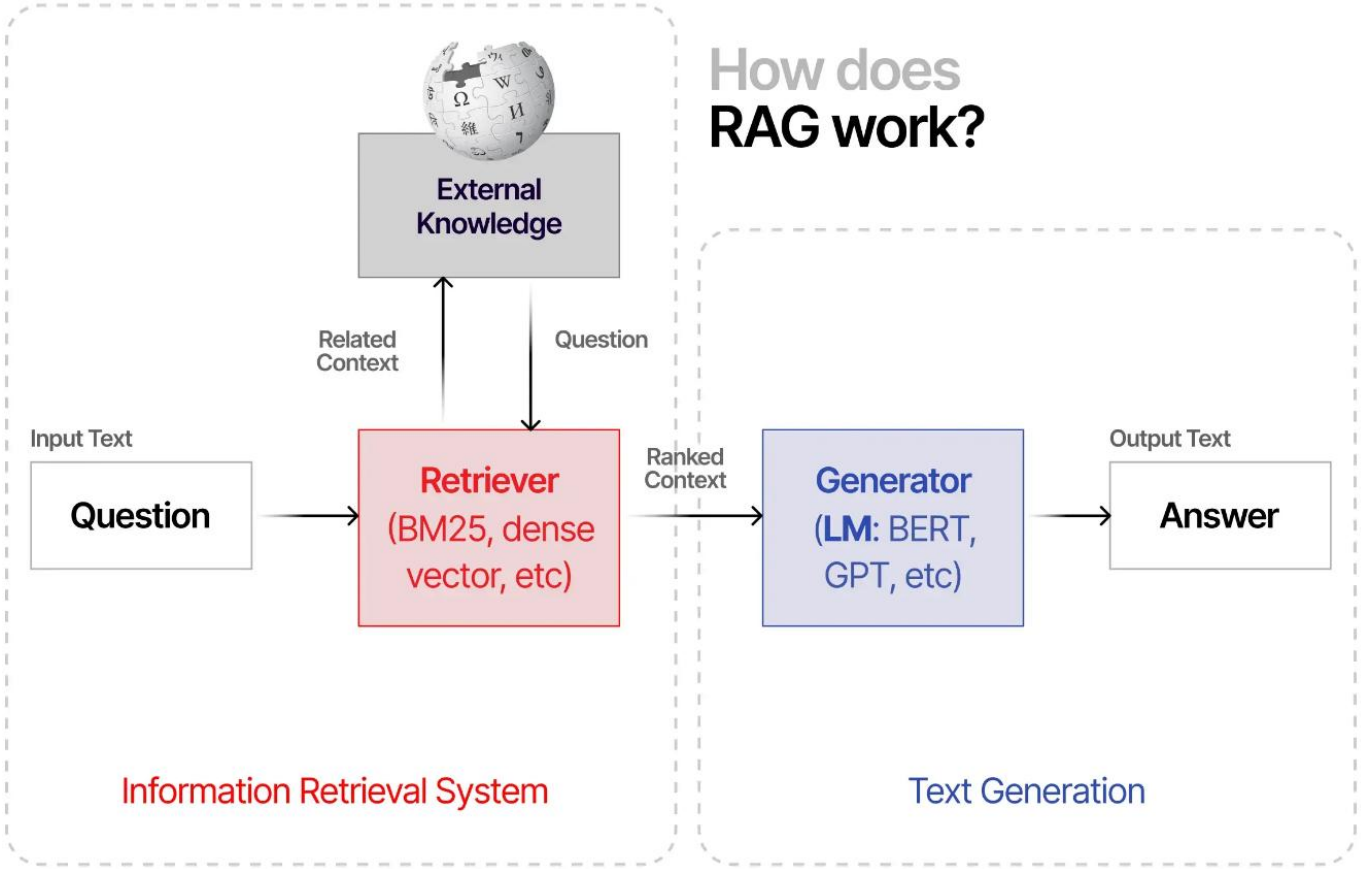


- A Transformer model is a
  - Neural network that learns context and thus meaning by tracking relationships in sequential data like the words in sentences.

- Transformer models apply an evolving set of mathematical techniques, called attention or self-attention,
  - to detect subtle ways even distant data elements in a series influence and depend on each other

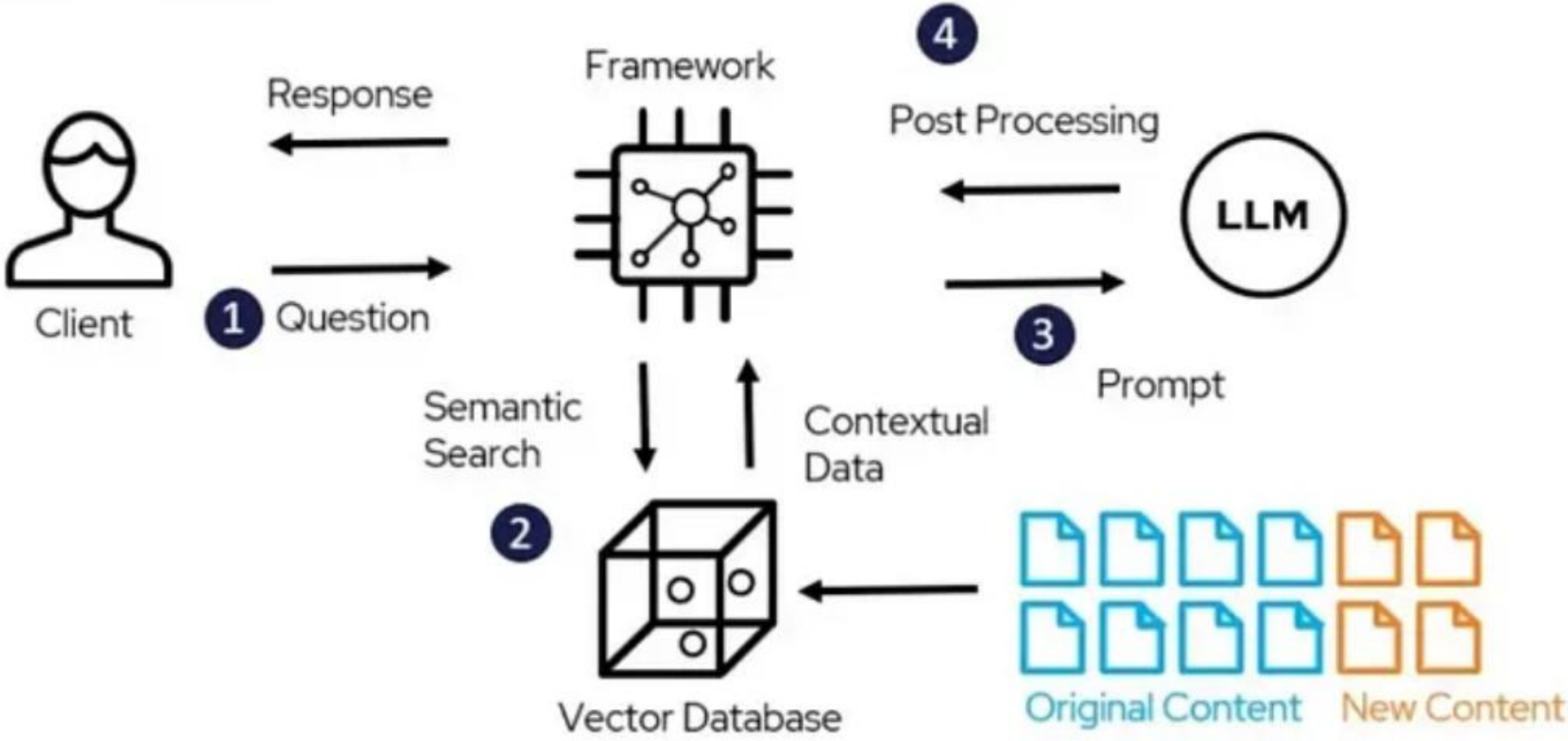
Large Language Models
Integrate Company Data

- Option 1: Fine tune the foundational model with your data.
- **Option 2:** Use Retrieval-Augmented Generation (RAG).



Large Language Models
Integrate Company Data

- Option 1: Fine tune the foundational model with your data.
- **Option 2:** Use Retrieval-Augmented Generation (RAG).



Big Data
Introduction

- No single standard definition.

WIKIPEDIA
The Free Encyclopedia

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Big data

62 languages

Contents hide

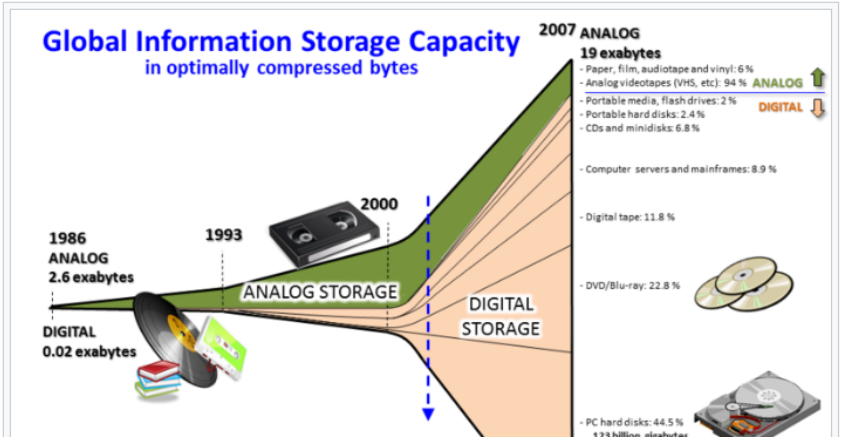
(Top)
- Definition
- Characteristics
- Architecture
- Technologies
- Applications
- Case studies
- Research activities
- Critique
- See also
- References

Article | Talk | Read | Edit | View history | Tools

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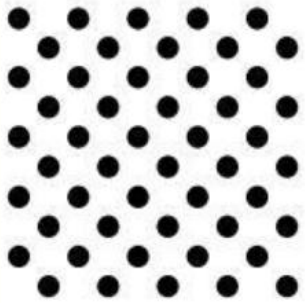
This article is about large collections of data. For the band, see Big Data (band). For the practice of buying and selling of personal and consumer data, see Surveillance capitalism.

Big data primarily refers to data sets that are too large or complex to be dealt with by traditional data-processing application software. Data with many entries (rows) offer greater statistical power, while data with higher complexity (more attributes or columns) may lead to a higher false discovery rate.[2] Though used sometimes loosely partly due to a lack of formal definition, the best interpretation is that it is a large body of information that cannot be comprehended when used in small amounts only.[3]



Big Data
The 3(5) Properties or Dimensions of Big Data

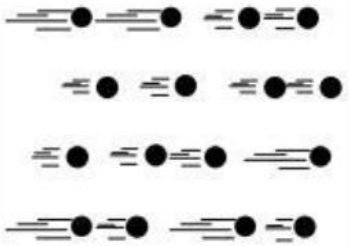
Volume



Data at Rest

Terabytes to exabytes of existing data to process

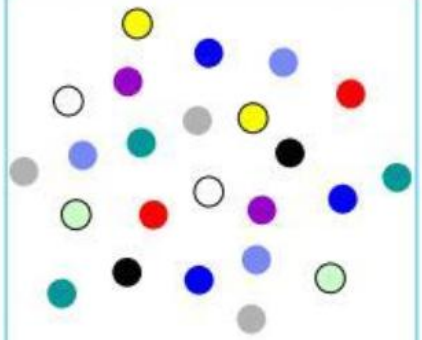
Velocity



Data in Motion

Streaming data, milliseconds to seconds to respond

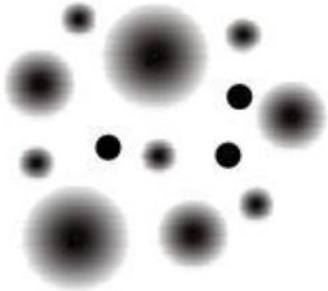
Variety



Data in Many Forms

Structured, unstructured, text, multimedia

Veracity*



Data in Doubt

Uncertainty due to data inconsistency & incompleteness, ambiguities, latency, deception, model approximations

Value*



Big Data
and it's Sizes

1. **Google:** processes 24 peta bytes of data per day.
2. **Facebook:** 10 million photos uploaded every hour.
3. **Youtube:** 1 hour of video uploaded every second.
4. **Twitter:** 400 million tweets per day.
5. **Astronomy:** Satellite data is in hundreds of PB.
6. ...
7. **"By 2020 the digital universe will reach 44 zettabytes..."**

Memory unit	Size	Binary size
kilobyte (kB/KB)	10^3	2^{10}
megabyte (MB)	10^6	2^{20}
gigabyte (GB)	10^9	2^{30}
terabyte (TB)	10^{12}	2^{40}
petabyte (PB)	10^{15}	2^{50}
exabyte (EB)	10^{18}	2^{60}
zettabyte (ZB)	10^{21}	2^{70}
yottabyte (YB)	10^{24}	2^{80}

The Digital Universe of Opportunities: Rich Data and the Increasing Value of the Internet of Things, April 2014.

That's 44 trillion gigabytes!

Big Data
Sources

Smarter Healthcare



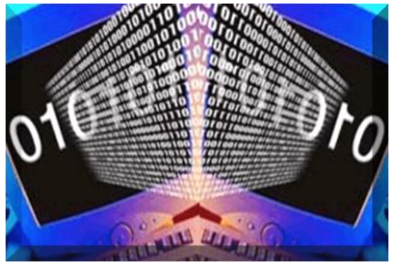
Multi-channel sales



Finance



Log Analysis



Homeland Security



Traffic Control



Telecom



Search Quality



Manufacturing



Trading Analytics



Fraud and Risk



Retail: Churn, NBO



Big Data
Sources

- Wherever we go, we are “datafied".
- Smartphones are tracking our locations.
- We leave a data trail in our web browsing.
- Interaction in social networks.
- Privacy is an important issue in Data Science.

Big Data
Introduction



Big Data
Enablers

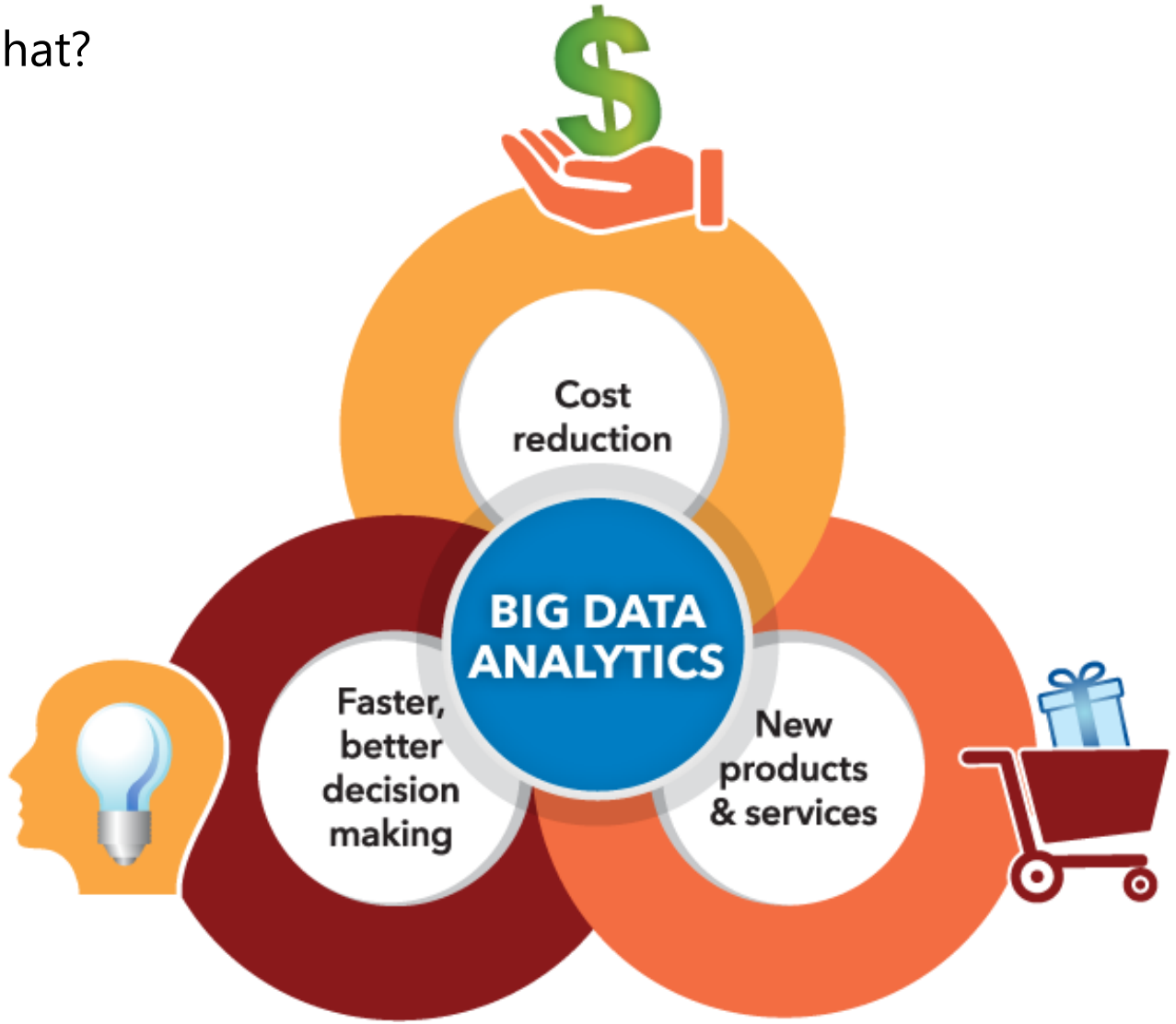
- Key enablers of appearance and growth of Big Data are
  - Increase of storage capacities
  - Increase of processing power
  - Availability of data

- Every day we create 2.5 quintillion bytes of data

- 90% of the data in the world today has been created in the last **two years** alone.

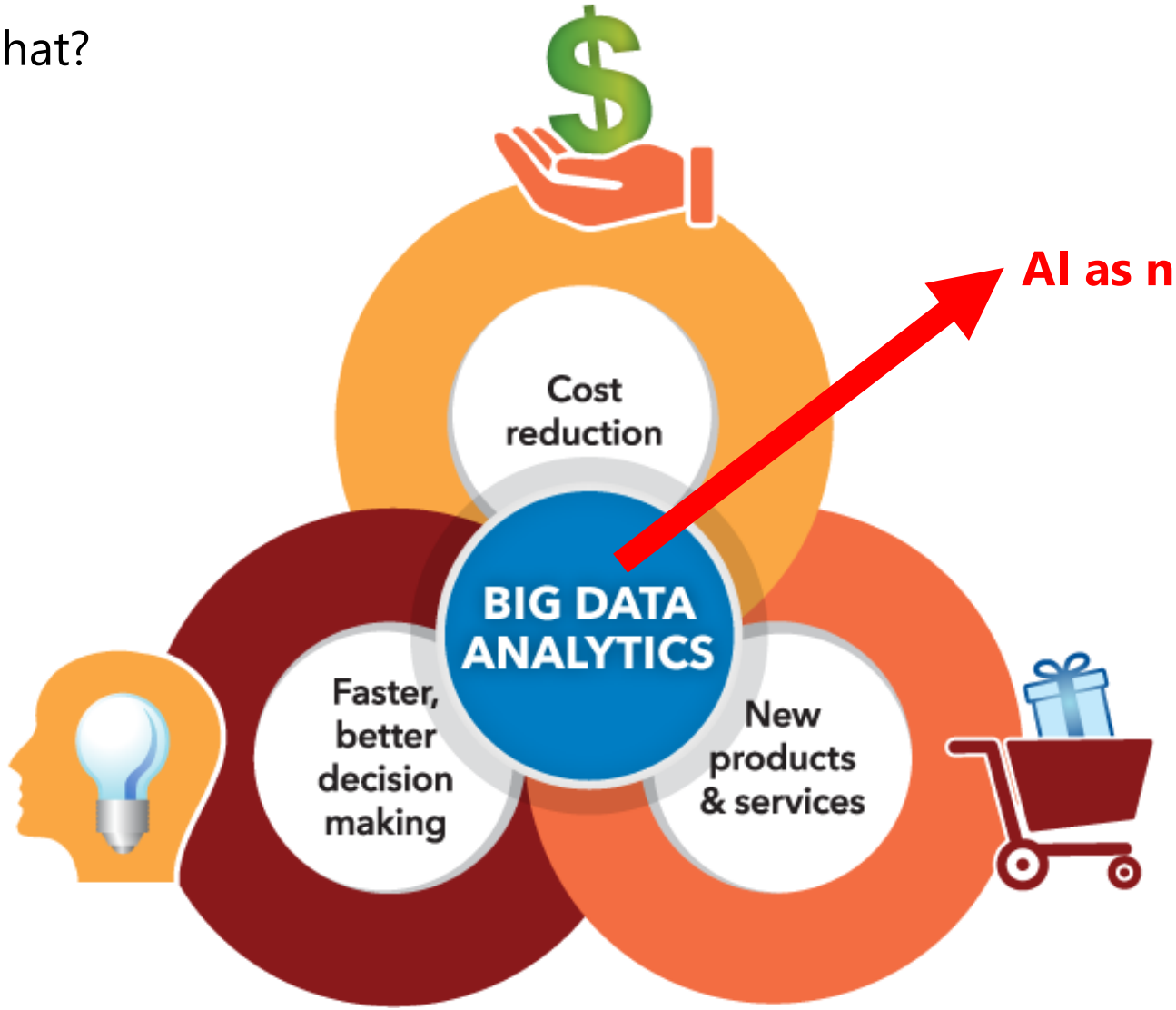
Big Data
Data Analytics

- Why we do all of that?



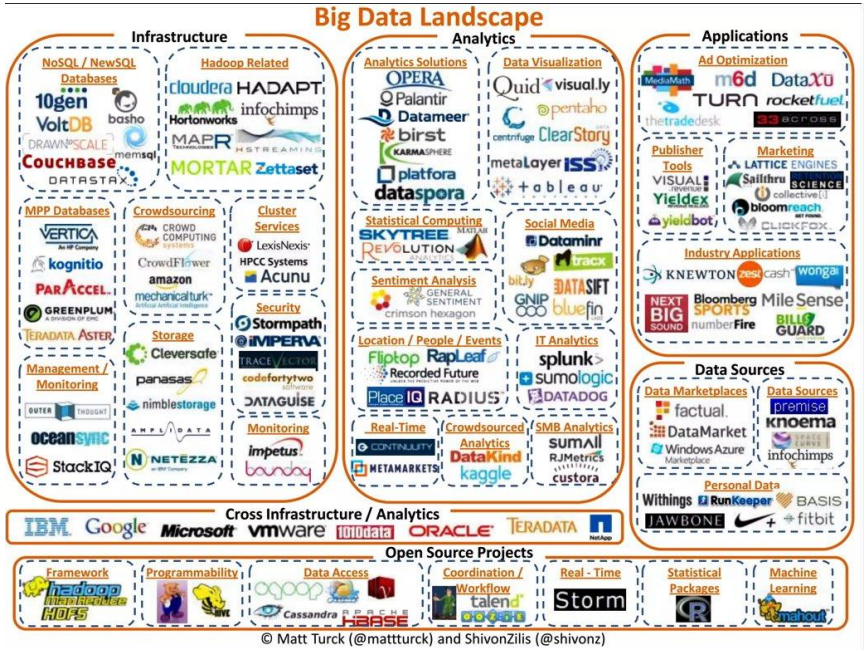
Big Data
Data Analytics

- Why we do all of that?

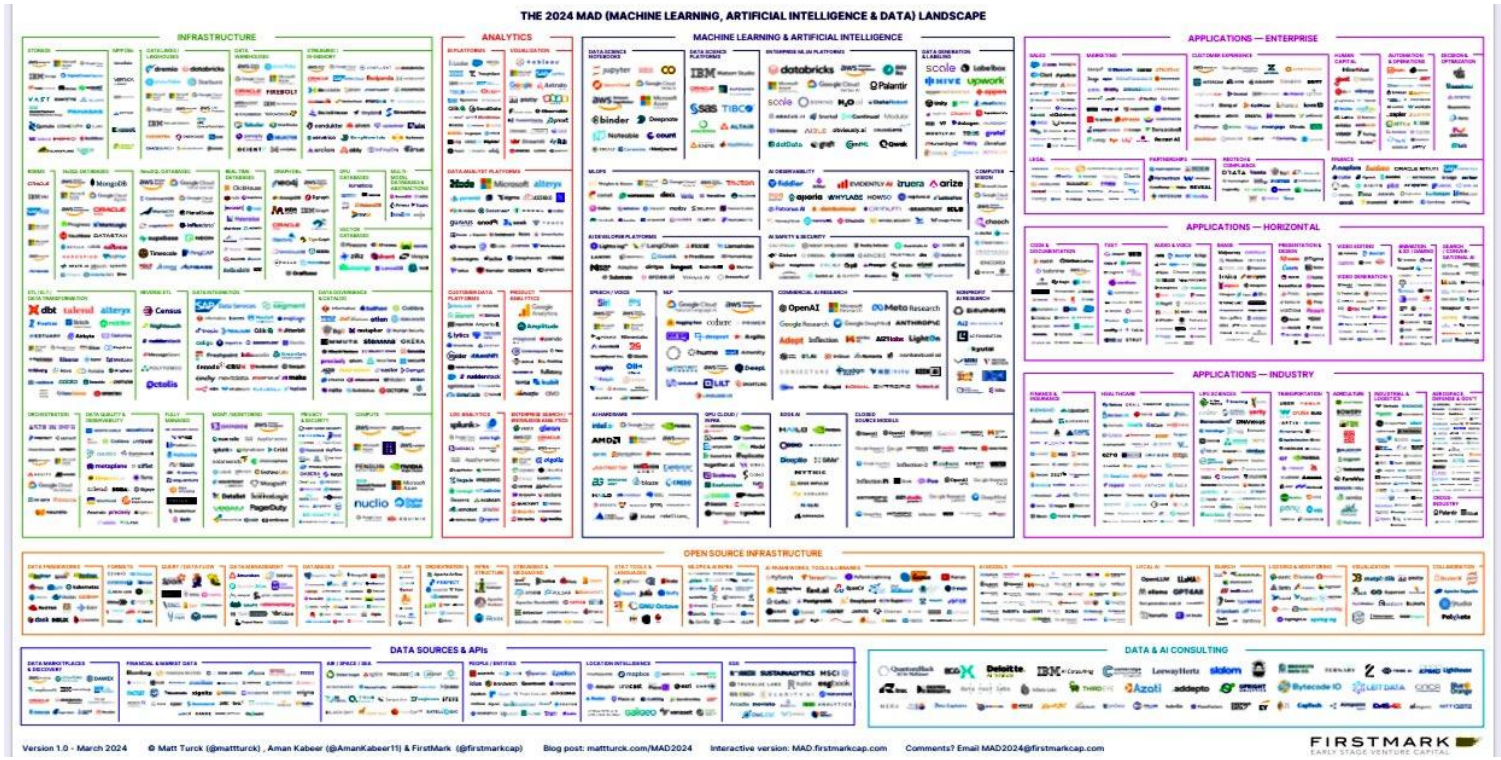


AI as new approach for analytics

Big Data
Landscape



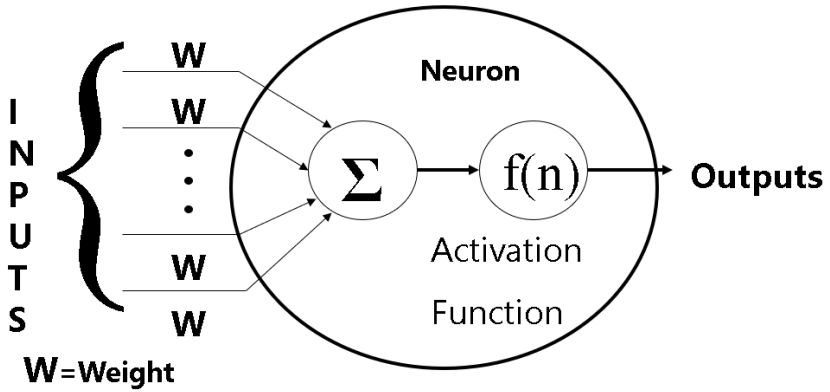
2012 (mainly warehousing)



2024 (mainly AI)

AI and Required Computation Power
Introduction

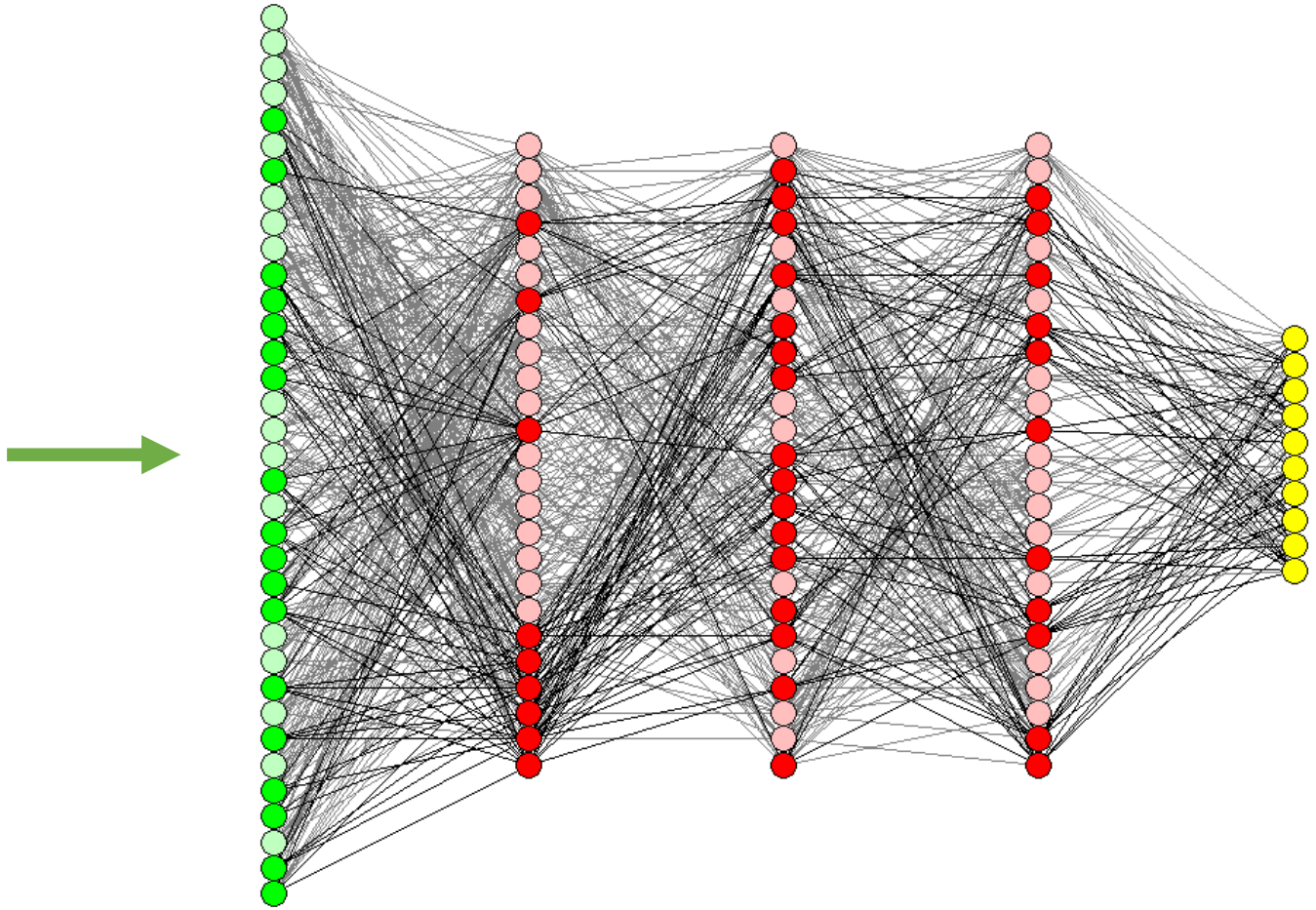
- As more data, as more accurate the **parameters** could be determined.
- Large amount of data require high computation time.
- The computation time is mainly influenced by the training phase (not the inference phase).



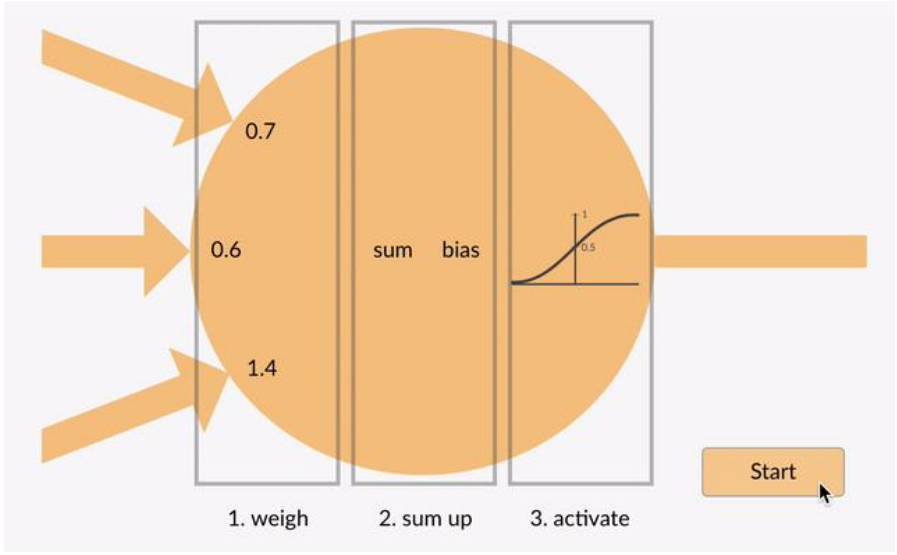
- During the training, we determine the „**weights**“ for each input for any neuron.

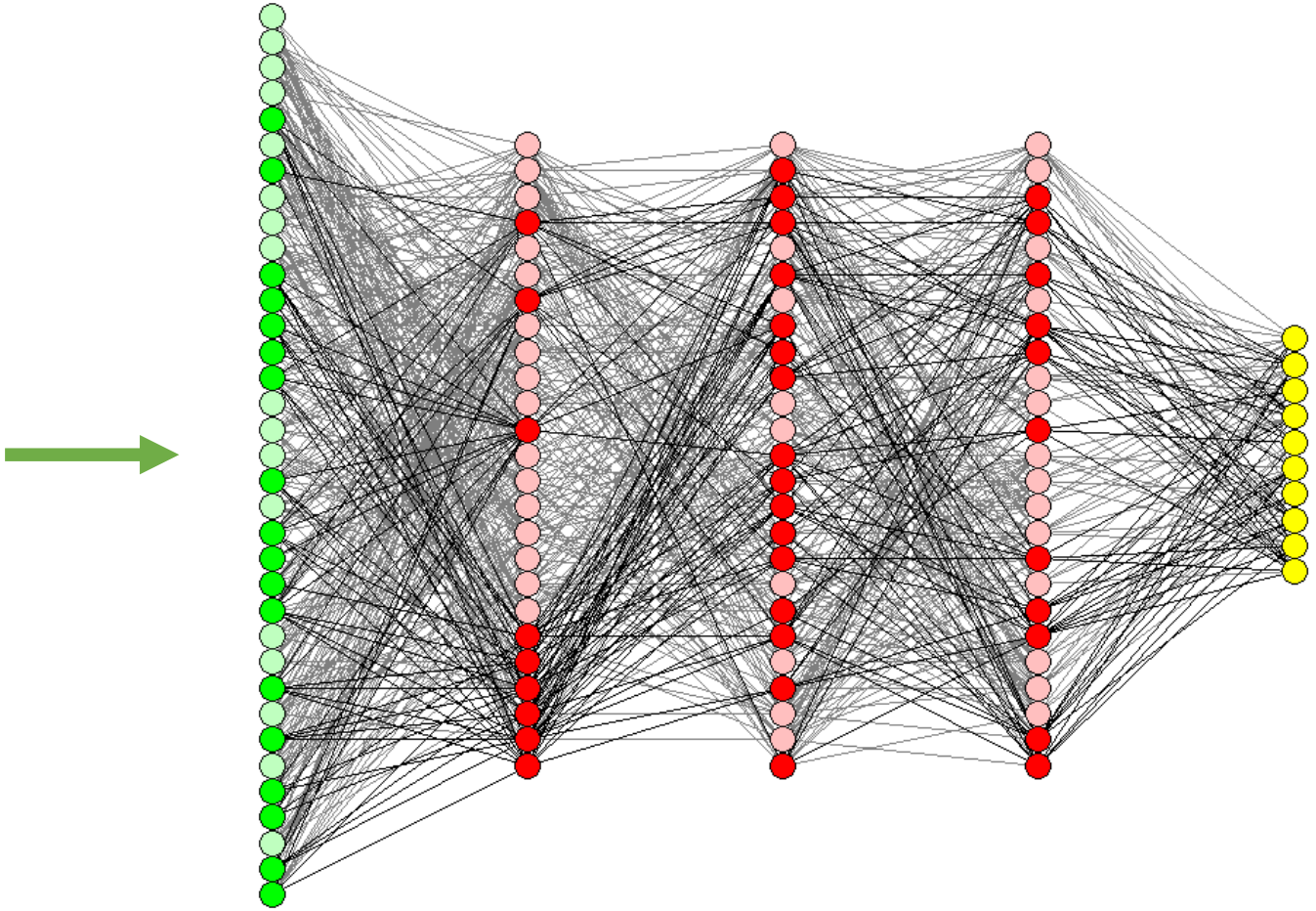
AI and Required Computation Power
Example

Period	Median					U.S. Average	
	U.S.	Northeast	Midwest	South	West	Houses Actually Sold	Constant-Quality House[3]
Annual Data							
1963	18,000	20,300	17,900	16,100	18,800	19,300	NA
1964	18,900	20,300	19,400	16,700	20,400	20,500	NA
1965	20,000	21,500	21,600	17,500	21,600	21,500	NA
1966	21,400	23,500	23,200	18,200	23,200	23,300	NA
1967	22,700	25,400	25,100	19,400	24,100	24,600	NA
1968	24,700	27,700	27,400	21,500	25,100	26,600	NA
1969	25,600	31,600	27,600	22,800	25,300	27,900	NA
1970	23,400	30,300	24,400	20,300	24,000	26,600	NA
1971	25,200	30,600	27,200	22,500	25,500	28,300	NA
1972	27,600	31,400	29,300	25,800	27,500	30,500	NA
1973	32,500	37,100	32,900	30,900	32,400	35,500	NA
1974	35,900	40,100	36,100	34,500	35,800	38,900	NA
1975	39,300	44,000	39,600	37,300	40,600	42,600	NA
1976	44,200	47,300	44,800	40,500	47,200	48,000	NA
1977	48,800	51,600	51,500	44,100	53,500	54,200	67,400
1978	55,700	58,100	59,200	50,300	61,300	62,500	77,400
1979	62,900	65,500	63,900	57,300	69,600	71,800	89,100
1980	64,600	69,500	63,400	59,600	72,300	76,400	98,100
1981	68,900	76,000	65,900	64,400	77,800	83,000	105,900
1982	69,300	78,200	68,900	66,100	75,000	83,900	108,400
1983	75,300	82,200	79,500	70,900	80,100	89,800	110,700
1984	79,900	88,600	85,400	72,000	87,300	97,600	115,100
1985	84,300	103,300	80,300	75,000	92,600	100,800	116,600
1986	92,000	125,000	88,300	80,200	95,700	111,900	121,200
1987	104,500	140,000	95,000	88,000	111,000	127,200	127,700
1988	112,500	149,000	101,600	92,000	126,500	138,300	132,400
1989	120,000	159,600	108,800	96,400	139,000	148,800	137,800
1990	122,900	159,000	107,900	99,000	147,500	149,800	140,400
1991	120,000	155,900	110,000	100,000	141,100	147,200	142,200
1992	121,500	169,000	115,600	105,500	130,400	144,100	144,100
1993	126,500	162,600	125,000	115,000	135,000	147,700	150,300
1994	130,000	169,000	132,900	116,900	140,400	154,500	157,500
1995	133,900	180,000	134,000	124,500	141,000	158,700	161,900
1996	140,000	186,000	138,000	126,200	153,900	166,400	165,100
1997	146,000	190,000	149,900	129,600	160,000	176,200	170,600
1998	151,600	200,000	156,000	134,900	163,000	181,300	175,000
Quarterly Data							
1997							
4th Quarter	144,200	195,000	144,500	129,000	159,000	175,400	171,100
1998							
1st Quarter	152,200	196,000	160,000	131,000	163,400	180,000	173,400
2nd Quarter	147,900	200,000	152,000	132,300	159,300	178,800	173,400
3rd Quarter	153,000	212,000	159,000	137,300	166,400	184,300	175,200
4th Quarter	150,400	198,000	149,100	134,500	164,900	179,400	177,300



AI and Required Computation Power
Example

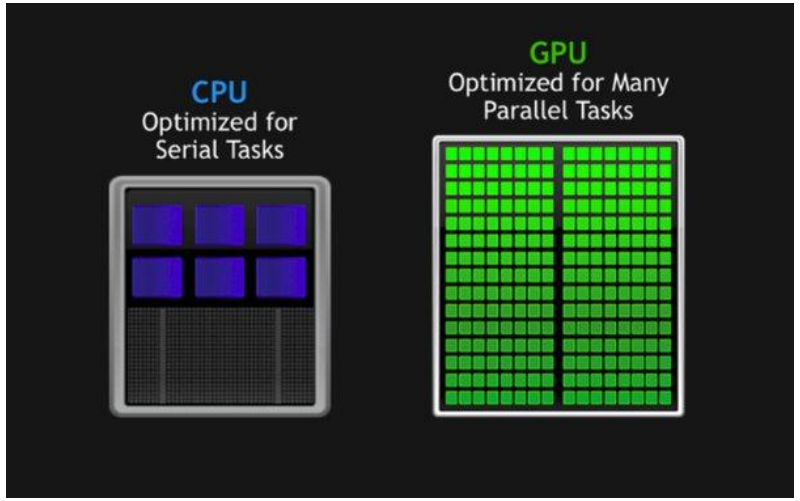




AI and Required Computation Power
Introduction

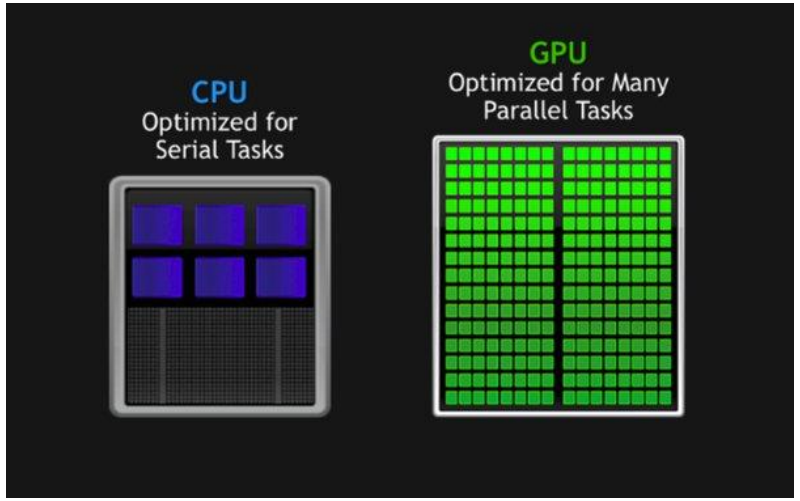
- GPUs and special hardware chips can help to reduce the computing time (not the effort).

- GPU are preferred because
  - Most operations are floating point operations (multiplication, divide, addition, subtraction, ..).
  - GPUs employ parallel processing (e.g. GeForce RTX 4090 contains 16384 cores with 10 threads per core)
  - Fast and large memory (e.g. GeForce RTX 4090 contains 24GB GDDR6)
  - Easy accessible technology since many frameworks exists.



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  - Easy accessible technology since many frameworks exists.
- But even GPUs are not powerful enough anymore…

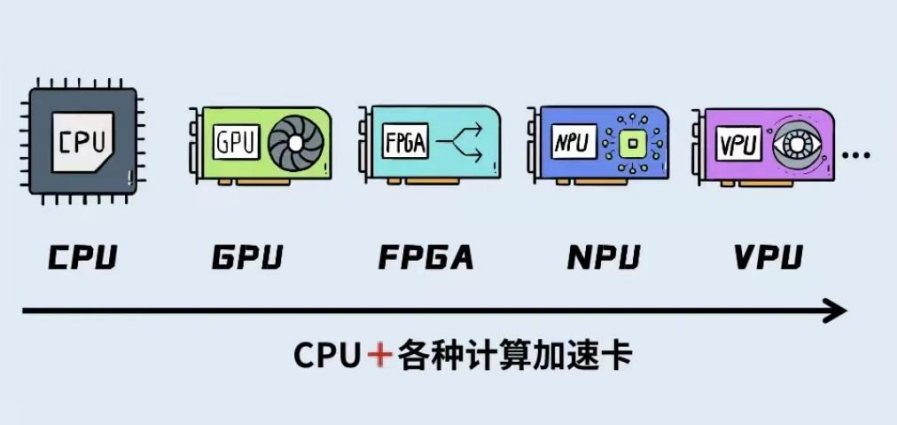


AI and Required Computation Power
Introduction

- Neural Processing Units (NPU).
- Also called Tensor Processing Units (TPUs).
- Designed specifically for the processing of neural network operations.
- NPUs are optimized for
    - high-volume (high throughput)
    - parallel computations
    - optimized for batch computation
    - multiplication and
    - activation function computation.

AI and Required Computation Power
Introduction

- Other companies are already working on next generation AI hardware.



CPU
- Small models
- Small datasets
- Useful for design space exploration

GPU
- Medium-to-large models, datasets
- Image, video processing
- Application on CUDA or OpenCL

TPU
- Matrix computations
- Dense vector processing
- No custom TensorFlow operations

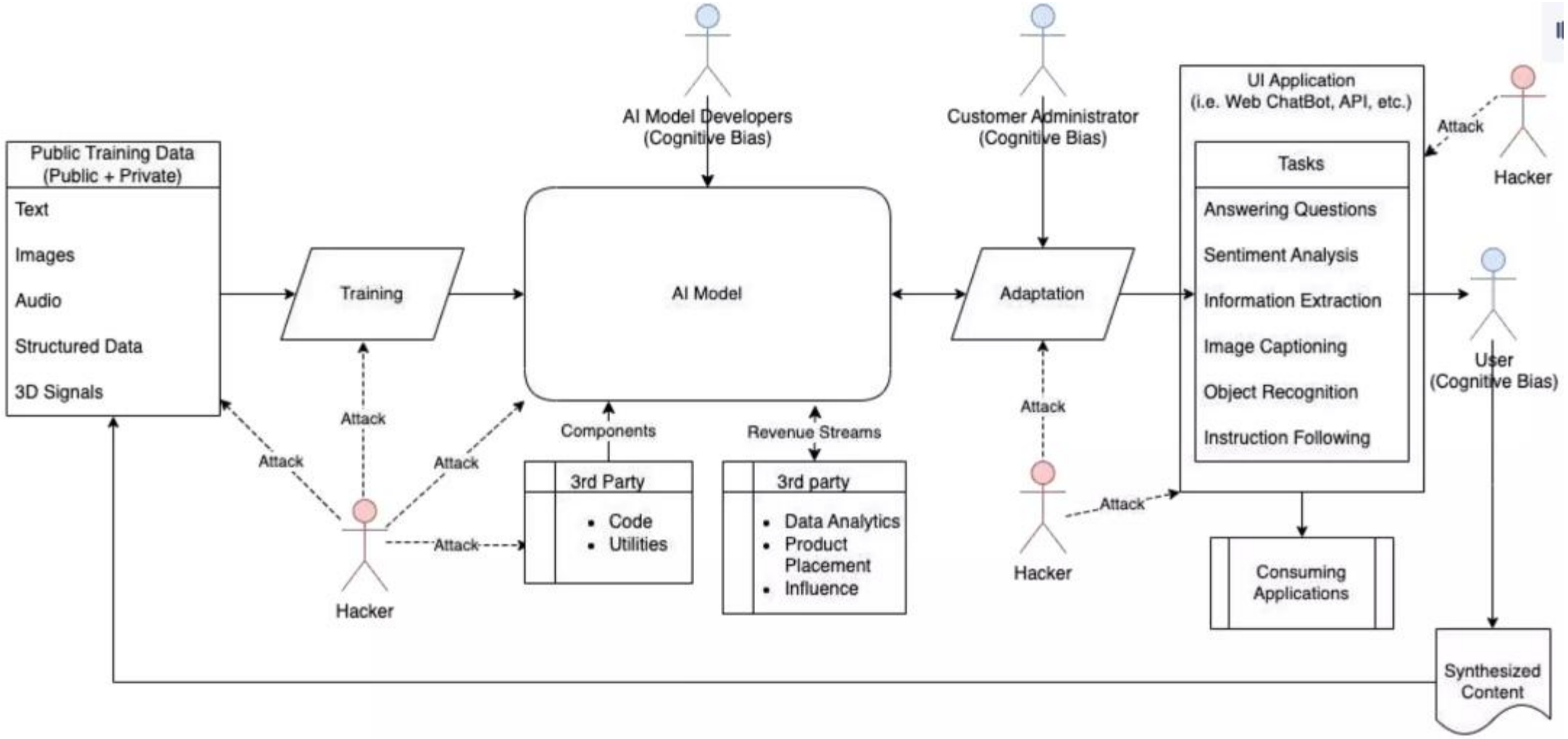
FPGA
- Large datasets, models
- Compute intensive applications
- High performance, high perf./cost ratio

- FPGA: field-programmable gate array (highly specialized for single operations)
- VPU: vision processing unit (highly specialized for machine vision operations)

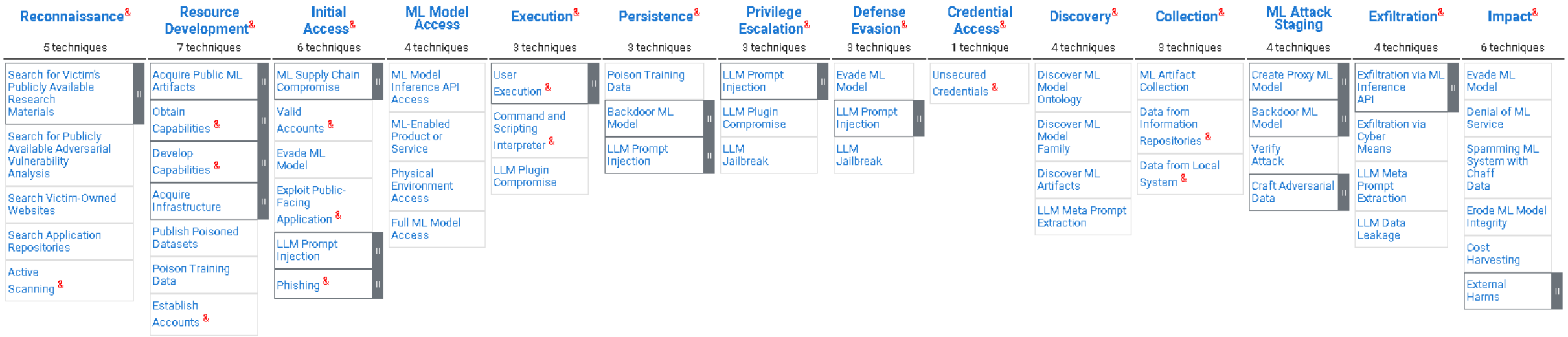
IT Security in AI Applications

- Same principals as in normal software have to be applied.
- But there is more…

IT Security in AI Applications
Potential Attach Surfaces



IT Security in AI Applications
Machine Learning Attack Chain



Columns: progression of tactics used in attacks.

Rows: ML techniques belonging to each tactic.

See Mitre Atlas Matrix https://atlas.mitre.org/matrices/ATLAS

IT Security in AI Applications
OWASP Top 10 2023

- For Machine Learning

ML01: Adversarial Attack

ML02: Data Poisoning Attack

ML03: Model Inversion Attack

ML04: Membership Inference Attack

ML05: Model Stealing

ML06: Corrupted Packages

ML07: Transfer Learning Attack

ML08: Model Skewing

ML09: Output Integrity Attack

ML10: Neural Net Reprogramming

Link to OWASP Top 10 ML

- For Large Language Model

LLM01: Prompt Injection

LLM02: Insecure Output Handling

LLM03: Training Data Poisoning

LLM04: Model Denial of Service

LLM05: Supply Chain Vulnerabilities

LLM06: Sensitive Information Disclosure

LLM07: Insecure Plugin Design

LLM08: Excessive Agency

LLM09: Overreliance

LLM10: Model Theft

Link to OWASP TOP 10 LLM

IT Security in AI Applications
Risk Similarities

OWASP 2023: Top 10 Machine Learning	OWASP 2023: Top 10 Large Language Models
ML01: Adversarial Attack	LLM01: Prompt Injection
ML02: Data Poisoning Attack	LLM03: Training Data Poisoning
ML09: Output Integrity Attack	LLM02: Insecure Output Handling
ML04: Membership Inference Attack	LLM06: Sensitive Information Disclosure
ML05: Model Stealing	LLM10: Model Theft
ML06: Corrupted Packages	LLM05: Supply Chain Vulnerabilities
ML03: Model Inversion Attack ML07: Transfer Learning Attack ML08: Model Skewing ML10: Neural Net Reprogramming	
	LLM04: Model Denial of Service LLM07: Insecure Plugin Design LLM08: Excessive Agency LLM09: Overreliance

IT Security in AI Applications
OWASP Top 10 for LLM

LLM01

Prompt Injection

This manipulates a large language model (LLM) through crafty inputs, causing unintended actions by the LLM. Direct injections overwrite system prompts, while indirect ones manipulate inputs from external sources.

LLM02

Insecure Output Handling

This vulnerability occurs when an LLM output is accepted without scrutiny, exposing backend systems. Misuse may lead to severe consequences like XSS, CSRF, SSRF, privilege escalation, or remote code execution.

LLM03

Training Data Poisoning

This occurs when LLM training data is tampered, introducing vulnerabilities or biases that compromise security, effectiveness, or ethical behavior. Sources include Common Crawl, WebText, OpenWebText, & books.

LLM04

Model Denial of Service

Attackers cause resource-heavy operations on LLMs, leading to service degradation or high costs. The vulnerability is magnified due to the resource-intensive nature of LLMs and unpredictability of user inputs.

LLM05

Supply Chain Vulnerabilities

LLM application lifecycle can be compromised by vulnerable components or services, leading to security attacks. Using third-party datasets, pre- trained models, and plugins can add vulnerabilities.

LLM06

Sensitive Information Disclosure

LLMs may inadvertently reveal confidential data in its responses, leading to unauthorized data access, privacy violations, and security breaches. It's crucial to implement data sanitization and strict user policies to mitigate this.

LLM07

Insecure Plugin Design

LLM plugins can have insecure inputs and insufficient access control. This lack of application control makes them easier to exploit and can result in consequences like remote code execution.

LLM08

Excessive Agency

LLM-based systems may undertake actions leading to unintended consequences. The issue arises from excessive functionality, permissions, or autonomy granted to the LLM-based systems.

LLM09

Overreliance

Systems or people overly depending on LLMs without oversight may face misinformation, miscommunication, legal issues, and security vulnerabilities due to incorrect or inappropriate content generated by LLMs.

LLM10

Model Theft

This involves unauthorized access, copying, or exfiltration of proprietary LLM models. The impact includes economic losses, compromised competitive advantage, and potential access to sensitive information.

Business Applications

- Spam filtering
- Credit card fraud detection
- Digit recognition on checks, zip codes
- Detecting faces in images
- MRI image analysis
- Recommendation system
- Search engines
- Handwriting recognition
- … and many more …

Costs and Profit Overview

golem.de
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MICROSOFT, GOLEM, OPENAI

KI-Tools bescheren Unternehmen bisher nur Verluste

Trotz des Hypes um KI-Systeme kämpfen Konzerne mit den enormen Kosten, die für Infrastruktur und Trainingszeiten aufgewendet werden.

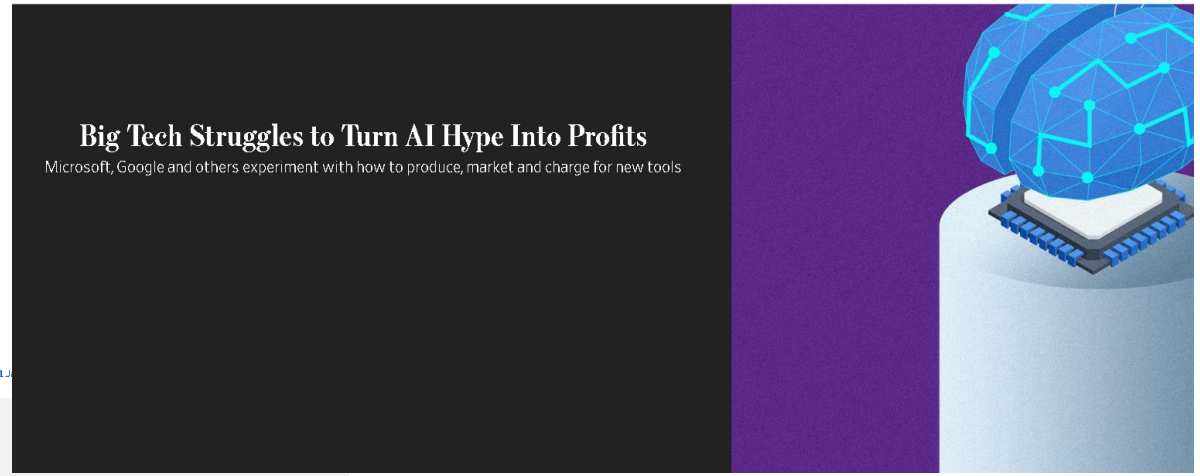
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10. Oktober 2023, 12:45 Uhr, Oliver Nickel

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Wall Street Journal: IT-Firmen wie Microsoft kämpfen mit den Kosten der KI

Das Wall Street Journal berichtet darüber, dass die großen IT-Firmen mit den von ihnen bereitgestellten KI-Tools aktuell kein Geld verdienen können.

Costs and Profit Overview

Meet the powerful tools everyone needs to thrive at work.
Learn how to create, edit, e-sign, and so much more with tools from Adobe Acrobat Pro. You can do it all anywhere, from an

NEWS

All ChatGPT users are getting GPT-4 for free with lifelike new model

Voices generated by the new GPT-4o model sound increasingly realistic.



By Mark Hachman
Senior Editor | MAY 13, 2024 12:33 PM PDT



AD GoD

Costs and Profit
Operation Costs at Microsoft

A single Nvidia A100 GPU can take up to 350 milliseconds to print a single word on ChatGPT. Chat-GPT is evolved from GPT-3.5, so there should be 175 billion parameters. The training data is about 4.5 billion TB. And a 3 billion parameter model can generate a token in about 6 milliseconds on NVIDIA A100 GPU (half precision+tensorRT+activation caching). Using this speed as a point of reference, a single Nvidia A100 GPU can take 350ms to print a single word on ChatGPT.

Since the latest version, ChatGPT-3.5, has more than 175 billion parameters, about five A100 GPUs are required to perform the necessary operations to obtain the output of a single query. After taking ChatGPT's ability to output about 15-20 words per second, an average of at least 8 A100 GPUs was obtained.

Each word generated on ChatGPT costs $0.0003. Considering Microsoft's current rate, a single Nvidia A100 GPU costs $3 per hour. According to the speed of generating a word in 350 milliseconds, the number of words that can be generated per hour can be calculated. Finally, the cost of each word generated on ChatGPT is 0.0003 Dollar.

Each query costs close to $0.01. Considering that chat-GPT needs about 30 words to reply to the user's query each time, it costs close to $0.01 per query. Three times higher than the cost of a Google search.

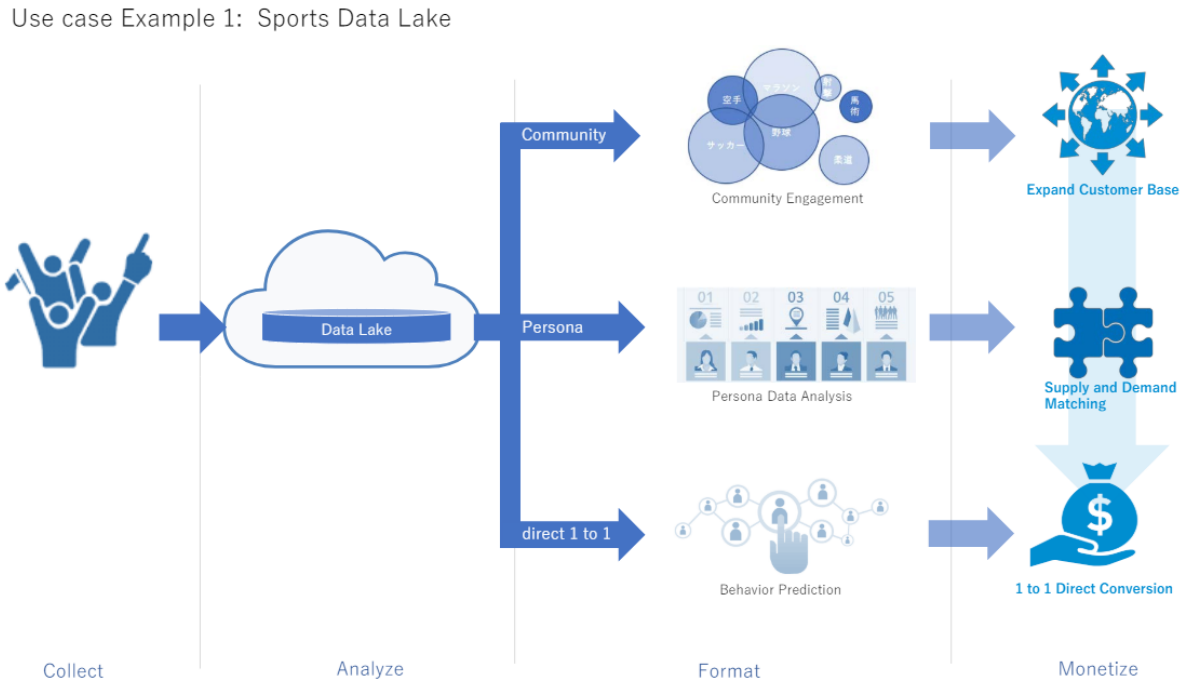
The daily operating cost is as high as $1.3 million. According to statistics from UBS, the number of monthly active users of Chat-GPT is expected to reach 100 million in January, and about 13 million people use it every day. Therefore, based on the calculation of 13 million people, each person queries 10 times a day, then there are 130 million queries per day, and the cost of each query is 0.01 US dollars, and the final daily cost is as high as 1.3 million US dollars.

Costs and Profit
Training Costs at Microsoft

A large language model such as ChatGPT requires at least thousands of GPU graphics clusters to train, and the construction cost itself is very high, and there is a certain limit on the number of calculations. After a certain number of calculations, resources will be exhausted. At this time, a new one needs to be built. GPU cluster.

In terms of the training cost of a single iteration, it costs 4.6 million U.S. dollars to train at least ten to dozens of times. Even if there are only 10 times, the cost will reach 46 million U.S. dollars.

How to Start an AI Project
Some Marketing Slides Recommend …



→ Start simple and tailor your use case very well.